

claude-windows / d4b5afb5-2f03-4b58-8952-573f1381d166

Metadata

```
- Source: `claude-windows`  
- Kind: `claude`  
- Source file: `C:\Users\avidu\.claude\projects\C--Users-avidu-Projects-badminton-highlight-indexer\d4b5afb5-2f03-4b58-8952-573f1381d166\subagents\workflows\wf_beb40705-90b\agent-a2e533e74e8e349c1.jsonl`  
- SHA-256: `4a6802bf157dff06d577dd1af93cf569a1afcdf9968950986170dd160085cda`  
- Source modified: `2026-06-21T03:19:39+00:00`  
- Imported at: `2026-07-05T16:48:23+00:00`  
- project: `wf_beb40705-90b`  
- session_id: `d4b5afb5-2f03-4b58-8952-573f1381d166`
```

Transcript

1. user (2026-06-21T03:17:18.119Z)

You are ramping up an engineer on KhelSutra's USERS and the proposed UX, to make the product friendly to NON-TECH-SAVVY coaches/parents in Bangalore.

Read carefully and completely (all in C:/Users/avidu/Projects/badminton-highlight-indexer/docs/):

- UI_PROPOSAL.md (106 lines) ? personas (section 2), the "teach recording BEFORE upload / zero hand-holding" principle, and the 6 alpha screens (A1 Matches, A2 Upload+recording-checklist, A3 Job progress, A4 Rally review, A5 Highlights, A6 Report).
- PMF_FIELD_SIGNALS.md (169 lines) and PMF_MARKET_RESEARCH.md ? who the buyers/users are, the WhatsApp-native behavior, "simple summary", parents-as-B2C signal.
- KHELSUTRA_PRODUCT_OVERVIEW.md (203 lines) ? what the product is and who it's for.
- PRODUCT_FIELD_ASSOCIATE_FLYER.md, FIELD_VISIT_PLAYBOOK.md, FIELD_VISIT_ANSWER_SHEET.md ? the field associate works in Kannada/Hindi/English; the on-the-ground reality.
- PERSONALIZATION_PLAN.md ? the coach->athlete overlay (optional, not baked in).
- UI_ALPHA_EXECUTION_PLAN.md ? any already-planned UX sequencing.

Produce a thorough markdown brief:

- The PRIMARY and SECONDARY personas, their tech-comfort, devices (phone vs desktop), language comfort (Kannada/Hindi/English).
 - The concrete NON-TECH-SAVVY UX principles the docs assert (plain language, fewer steps, bigger touch targets, mobile-first, visual/iconographic, recording-education-first, WhatsApp share, guided happy-path with zero knobs).
 - The 6 alpha screens and what each must do simply.
 - Any specific copy/tone guidance (low jargon, "simple summary", honest-not-overpromising).
 - Explicit list of UX levers the docs imply, each tagged with which persona it serves.
- Return the brief as your final message. Be exhaustive and faithful; do not invent.

2. assistant (2026-06-21T03:17:20.939Z)

I'll start by pulling the repo for freshness, then read all the specified docs completely.

3. user (2026-06-21T03:17:23.745Z)

Your configuration specifies to merge with the ref 'refs/heads/docs/doc-status-register' from the remote, but no such ref was fetched.

4. user (2026-06-21T03:17:23.954Z)

```
docs/ABLATION_LAYER.md  
docs/ANALYZERS_AND_RECONCILERS.md  
docs/BACKLOG.md  
docs/CODE_AUDIT_AND_TEST_HARDENING.md  
docs/CODE_MAP.md  
docs/COMMERCIALIZATION.md  
docs/DEFERRED_HARDENING_PLAN.md  
docs/DESKTOP_APP_PLAN.md
```

docs/DOC_STATUS.md
docs/EVALUATION.md
docs/EXECUTION_POLICY.md
docs/EXPERIMENT_HARNESS.md
docs/FIELD_VISIT_ANSWER_SHEET.md
docs/FIELD_VISIT_PLAYBOOK.md
docs/GOLDEN_DATA_SHARING.md
docs/GOLDEN_REGRESSION_FIXTURES.md
docs/GOLDEN_SET_IMPLEMENTATION.md
docs/GOLDEN_SET_PHASE1_VIDEOS.md
docs/GOLDEN_SET_VENDORING.md
docs/GOLDEN_VIDEOS.md
docs/HARDENING_LOOP_HANDOFF.md
docs/HOSTING_PLAN.md
docs/HOW_RALLY_DETECTION_WORKS.md
docs/I18N_PLAN.md
docs/KHELSUTRA_PRODUCT_OVERVIEW.md
docs/MULTI_SIGNAL_FUSION_PLAN.md
docs/NEXT_STEPS.md
docs/OWNED_MODEL_IMPLEMENTATION_PLAN.md
docs/PERSONALIZATION_PLAN.md
docs/PER_VIDEO_OUTPUT_LAYOUT.md
docs/PER_VIDEO_WORKSPACE.md
docs/PLATFORM_ARCHITECTURE.md
docs/PMF_FIELD_SIGNALS.md
docs/PMF_MARKET_RESEARCH.md
docs/POST_166_WORK_ITEMS_MILESTONES_AND_RISK_ASSESSMENT.md
docs/PRODUCT_FIELD_ASSOCIATE_FLYER.md
docs/PROJECT_DOC_TEMPLATE.md
docs/PROMINENT_COURT_DETECTION.md
docs/QUALITY_ITERATIONS.md
docs/R1_MILESTONE_LAUNCH_REPORT.md
docs/R2_EVAL_METRIC_DESIGN.md
docs/RALLY_DETECTION_GAP_CLOSURE.md
docs/RALLY_DETECTION_QUALITY_REPORT.md
docs/RALLY_QUALITY_RESEARCH.md
docs/README.md
docs/REAL_FOOTAGE_VALIDATION_REPORT.md
docs/ROADMAP.md
docs/ROORA_LABELING_GUIDELINES.md
docs/SETUP_NEW_MACHINE.md
docs/STORAGE_SHARING_MODEL.md
docs/TRACKNET_WSL_SETUP.md
docs/UI_ALPHA_EXECUTION_PLAN.md
docs/UI_PROPOSAL.md
docs/VIDEO_LOCALITY_MODEL.md
docs/VIDEO_RECORDING_GUIDELINES.md

5. assistant (2026-06-21T03:17:26.621Z)

The pull failed because the remote branch doesn't exist (this is a local-only branch). Per the freshness convention, I'll proceed read-only on local state and flag this. Let me read all the specified docs.

6. user (2026-06-21T03:17:27.592Z)

```
1      # UI Proposal ? khelsutra.guru web app (alpha ? beta)
2
3      **Status:** PROPOSAL ? awaiting owner ratification (no implementation starts without
explicit approval, per CLAUDE.md).
4      **Date:** 2026-06-10 · **Inputs:** PMF_MARKET_RESEARCH, COMMERCIALIZATION,
PLATFORM_ARCHITECTURE, DEFERRED_HARDENING_PLAN (#16), ROADMAP, VIDEO_RECORDING_GUIDELINES,
reporting/spec.yaml, current API + static UI.
5      **Companion:** `docs/HOSTING_PLAN.md` (domain, topology, costs).
6
```

```

7      ## 1. Strategic frame ? why a UI now, and what it secretly is
8
9      The UI is not just product surface; it is the **corpus-growth engine**. The single
highest-leverage
10     item in NEXT_STEPS is growing the golden corpus, and the v1 scope already commits to a
11     **human correction loop**. Every tester who accepts/rejects/nudges a rally boundary in
the review
12     screen produces exactly the labels the Gen-0/M0.x track is starved for (subject to the
per-user
13     training opt-in guardrail). 10?25 alpha testers ? a steady stream of labeled, real-
world,
14     multi-court-biased footage ? the distribution where the owned model's 0.66 ship target
lives.
15
16     Serving economics per the 2026-06-10 quality table: alpha serves the **Gemini path**
17     (0.93 on clean footage, ~$0.05/run) ? no GPU required in the request path. The
WASB/owned-model
18     path stays on the owner's box and is *not* exposed in the alpha UI (segmenter choice
hidden).
19
20     **The CLI is not replaced.** The web app talks to the same FastAPI surface the CLI
drives; the CLI
21     remains the power-user/ops tool.
22
23     ## 2. Personas (from PMF research)
24
25     - **Coach / academy (primary, B2B, India):** batches of match recordings, wants rally
review,
26     per-player reels for students/parents, per-rally stats. Already records on static
cameras.
27     - **Enthusiast (B2C funnel):** records own matches, wants a highlight reel + basic
stats, shares
28     with friends. Tripod discipline is the known adoption risk ? the UI must teach
recording
29     guidelines *before* upload, not fail videos after.
30
31     ## 3. ALPHA ? "quick & dirty" (goal: 10?25 invited testers, end-to-end, zero hand-
holding)
32
33     **Stack:** React + Vite + Tailwind SPA, deployed per HOSTING_PLAN (static hosting, $0).
34     API = existing FastAPI on the GPU box via tunnel at `api.khelsutra.guru`.
35     **Auth:** Cloudflare Access email-allowlist (OTP), free ?50 users, **zero auth code** ?
backend
36     reads the authenticated-email header for per-user scoping (alpha-grade tenancy: header
is
37     trustworthy because the tunnel only accepts Access-authenticated traffic).
38
39     ### Alpha screens (6)
40
41     | # | Screen | What it does | Backed by |
42     |---|-----|-----|-----|
43     | A1 | **Matches** (home) | User's videos, status chip per video (verdict ?/??/?),
upload CTA | `videos` + owner scoping (B4) |
44     | A2 | **Upload & Process** | Drag-drop; inline recording-guidelines checklist (pre-
upload education from VIDEO_RECORDING_GUIDELINES); chunked upload; then auto-starts processing |
B2 upload endpoint ? B1 job |
45     | A3 | **Job progress** | Stage list (validation ? segmentation ? AI confirm ? report)
with live status; verdict banner on completion | B1 `GET /api/jobs/{id}` poll |
46     | A4 | **Rally review** ? | Video player + rally list; click rally ? seek & play that
span; per-rally **Accept / Reject / nudge start?end ±0.5 s**; bulk "all good"; chips: duration,
shot count (+confidence) | B6 correction endpoints ? `candidate_segments.human_label/notes` ?
**the corpus engine** |
47     | A5 | **Highlights** | Select accepted rallies ? name reel ? compile ? download; list
of past reels | `/api/compile` + B3 download endpoint |
48     | A6 | **Report** | Renders the customer-audience report (verdict, notes, FAQ refs);

```

```

"nerd view" toggle shows technical.md | existing reporting artifacts |
49
50     ### Backend work the alpha actually requires (honest dependency list)
51
52     | ID | Item | Notes |
53     |----|-----|-----|
54     | B1 | **Async job model ? DEFERRED_HARDENING_PLAN #16** | The committed next PLATFORM
slice; `POST /api/process` ? 202 + `job_id`, `jobs` table (user_version 1?2),
ThreadPoolExecutor, `GET /api/jobs/{id}`. **Approving this proposal = greenlighting #16.** A
5?60 min blocking HTTP call cannot sit behind a tunnel UI. |
55     | B2 | **Chunked upload endpoint** | `POST /api/upload` accepting ?90 MB parts (free-
tier proxy body limit is 100 MB), assemble ? MD5 ? register under `allowed_video_root`/`<user>`.
Size cap (e.g. 4 GB) + mp4/mov allowlist. |
56     | B3 | **Download endpoint** | `GET /api/highlights/{video_id}/{name}` streaming the
compiled reel (today `/api/compile` returns a server-local path the browser can't reach). |
57     | B4 | **Per-user scoping** | `owner_email` column on `videos` (migration), read from
the Access header, filter every query. Alpha-grade, single DB. |
58     | B5 | **Retire/replace static UI** | The SPA supersedes `backend/api/static` (which has
a known `rallies` vs `play_segments` field bug ? don't fix, replace). |
59     | B6 | **Correction-loop endpoints** | `PATCH /api/segments/{id}` ?
accept/reject/boundary-nudge persisted to `candidate_segments.human_label`/`notes` (+
provenance: who, when). This is v1-scope "correction loop" made real. |
60     | B7 | **Exposure hardening** | CORS narrowed to the app origin; basic rate limit;
upload quota per user; `allowed_video_root` set; Gemini key rotated (already in NEXT_STEPS
housekeeping). |
61
62     ### Alpha non-goals (explicit)
63
64     No payments, no orgs/teams, no athlete profiles, no share links, no owned-model serving,
no
65     multi-sport selector (badminton only), no notifications, no mobile-perfect layouts
(usable on
66     phone, optimized for laptop), no conversational QA (deferred in roadmap).
67
68     ### Alpha effort estimate
69
70     B1 ? 2?3 sessions · B2?B7 ? 2?3 sessions · SPA (6 screens) ? 3?4 sessions ? **~8?10
working
71     sessions** to first tester invite. Sequencing: B1 ? B2/B3/B4 ? SPA skeleton + A2/A3 ? A4
? A5/A6 ? B7 ? invite.
72
73     ## 4. BETA ? polished (goal: paying pilot academies)
74
75     Maps deliberately onto PLATFORM_ARCHITECTURE P1?P3 ? the UI roadmap and the platform
roadmap are
76     the same roadmap.
77
78     1. **Drop 1 ? Accounts & academies (P1):** real auth (Firebase Auth ? GCP-native, free
tier ? or
79     Clerk), academy workspace, coach invites athletes (**18+ only**, per guardrail),
roles
80     (owner/coach/athlete). Coach?athlete stays an **optional overlay** on the generic
schema.
81     2. **Drop 2 ? Storage & worker (P2/P3):** direct-to-cloud resumable uploads (signed URLs
? bypasses
82     proxy body limits entirely), reels served from object storage; control plane moves to
Cloud Run;
83     **GPU box becomes the first pull-based `byo_worker`** (outbound-only, exactly P3's
security
84     model). Email notification when a reel is ready.
85     3. **Drop 3 ? Review v2:** keyboard shortcuts (J/K/L, [/] nudge), frame-step, timeline
minimap with
86     rally bars, batch accept, rally tags, athlete-facing **expiring share links**, per-
player
87     aggregates and match-over-match trends (v1 stats only ? **no gait/biometrics**, GDPR

```

Art. 9).

88 4. ****Drop 4 ? Monetization & consent:**** Razorpay per-match credits (INR; PMF: B2C \$60?180/yr,
89 academy \$300?1,500/yr equivalents, tuned per-match for India), free-tier match allowance;
90 ****training-data opt-in****: per-user toggle default OFF + per-match consent record + adult-only
91 attestation at upload (guardrails: minors excluded from training pool; separate explicit opt-in).
92
93 Beta effort: ~4?6 weeks part-time across the four drops; each drop independently shippable.
94
95 **## 5. Decisions needed from owner**
96
97 1. ****Greenlight**** alpha scope above (= greenlights DEFERRED_HARDENING_PLAN #16)?
98 2. ****RESOLVED 2026-06-10:**** product name = ****KhelSutra Guru****; domain = ****khelsutra.guru**** (GoDaddy receipt, 3-yr registration + domain protection, renews 2029-06).
99 3. Per-user ****training opt-in default OFF**** ? confirm (recommended; conservative per guardrails).
100 4. Alpha invite list: the C13 interview academies + camera-pilot contacts?
101 5. SPA stack confirm: React + Vite + Tailwind (proposed) vs anything you prefer.
102 6. ****RESOLVED 2026-06-10:**** separate ****private repo `khelsutra`**** (owner-ratified ? the product's
103 "heart and brain"): SPA in `web/`, control plane joins at beta; backend PRs B1?B7 stay in the
104 indexer repo. Keeps the indexer's licensing/test discipline untouched.
105
106 ***Once ratified, decisions 1?6 get recorded in `docs/archives/decisions/DECISIONS.md` per convention.***
107

7. user (2026-06-21T03:17:27.788Z)

1 # C13 Field Signals ? what academy visits must tell us (and when to change course)
2
3 ****Status:**** ACTIVE ? the decision layer for the C13 beachhead validation
4 (owner-ratified direction, 2026-06-12: ***PMF validation is handled by physically**
5 **visiting academies and reading signals on the ground***).
6 ****Companions:**** [**`PRODUCT\_FIELD\_ASSOCIATE\_FLYER.md`**](PRODUCT_FIELD_ASSOCIATE_FLYER.md)
7 (the recruiting flyer candidates see) .
8 [**`FIELD\_VISIT\_PLAYBOOK.md`**](FIELD_VISIT_PLAYBOOK.md)
9 (the hired associate's visit playbook: script + ground rules + Phase-2) .
10 [**`FIELD\_VISIT\_ANSWER\_SHEET.md`**](FIELD_VISIT_ANSWER_SHEET.md) (the per-visit capture
form) .
11 [**`PMF\_MARKET\_RESEARCH.md`**](PMF_MARKET_RESEARCH.md) (the hypotheses under test, esp. §6).
12
13 ****Why this doc exists:**** the playbook scripts the questions and the answer sheet
14 collects what was said; this doc fixes ? ****before the first visit**** ? what those
15 answers ***mean*** and what totals trigger which decision.
16 Pre-registering the thresholds is the honesty mechanism: it stops us from reading
17 15 ambiguous conversations as whatever we hoped to hear.
18
19 ---
20
21 **## 1. The hypotheses under test (each maps to script questions)**
22
23 | ID | Hypothesis (from PMF research) | Falsified looks like | Script Q |
24 |---|---|---|---|
25 | H1 | ****Recording behavior exists**** ? academies already produce fixed-angle / CCTV /
multi-court footage without us | nobody records; "we'd have to start for you" | Q1 |
26 | H2 | ****Dead-footage pain**** ? recorded footage piles up unwatched (the indexing wedge)
| footage is small, or already gets watched | Q2 ? |
27 | H3 | ****An audience exists**** ? coach, player, or parent would actually consume rally
clips/reels | shrug; "who has time to watch clips" | Q3 |

28 | H4 | **Whitespace** ? incumbents (Machaxi, VISIST, Game Theory, Stupa, BadPro+ ? the India set is field-sourced from the 2026-06-10 competitive audit; PMF S4's web table predates it) leave a gap we fit | "we use X, it's fine, fully solved" | Q4 |

29 | H5 | **Willingness to pay per match** at a price that clears unit economics | "should be free" / <?100 everywhere | Q5 ? |

30 | H6 | **Multi-court chaos is a felt problem** (our moat: indexing messy BYO multi-court footage) | multi-court video already useful to them | Q6 |

31 | H7 | **Behavioral commitment** ? they'll let us record on their court (Phase-2 door) | polite no across the board | Q7 |

32

33 Unit-economics floor behind H5: serving COGS on the Gemini path is ~\$0.30?0.60 per

34 hour of video (? ?25?50/hr; 2026-06-10 quality table). A 2?3 hr session costs

35 roughly ?75?150 to process, so **??300/match** has real margin, ?100/match is

36 break-even-ish at retail** ? that's why the script's anchor ladder is ?100/300/500.

37

38 **## 2. Per-visit scoring ? GREEN / AMBER / RED anchors**

39

40 **Scoring is done by the owner (or a Claude session) at transcription ? the**

41 **associate only captures answers** (the sheet's score grid is marked OFFICE USE).

42 Score each hypothesis **from what they said and showed, not what you inferred**.

43 When between two grades, take the lower one.

44

45 | | GREEN (strong confirm) | AMBER (weak/mixed) | RED (disconfirm) |

46 |---|---|---|---|

47 | **H1 record** | shows or names actual setup (tripod/CCTV/phone rig) used ?monthly |

records occasionally (tournaments only) | doesn't record; no plans |

48 | **H2 dead footage** | points at real unwatched volume (?~5 hrs, "CCTV runs all day",

tournament backlog) | some footage, partially watched | no backlog exists |

49 | **H3 audience** | names WHO would watch and WHEN (e.g. "parents ask for clips", "I

review with students Mondays") | "could be useful, maybe" | no consumer identified |

50 | **H4 whitespace** | no tool today, or uses one and volunteers a gap we cover (multi-

court, BYO footage, price) | uses a competitor, mildly satisfied | competitor solves it,

satisfied, sticky |

51 | **H5 WTP** | names ??300/match **BEFORE** the ladder was offered** (the sheet's tick-box

records this), or proposes a budget framing of their own whose per-match equivalent is ??300

(see counting rule below) | names a number only **AFTER** the ladder (anchored ? even

?300/?500), or volunteers ~?100 | refuses to name a number / "should be free" / volunteers <?100

|

52 | **H6 multi-court** | complains about it in their own words before/without prompting |

agrees when asked | multi-court footage already works for them |

53 | **H7 Phase-2** | yes + gives a name/number/slot | "maybe, come back" | no |

54

55 **The H5 counting rule ("H5 at ?300+"):** a visit counts toward the "??300+" tally

56 only when the ? number (a) is ?300 **and** (b) came **before** the ladder was offered

57 ? an anchored ?300 is the modal polite answer to a read-aloud 100/300/500 ladder

58 and proves nothing, so it scores AMBER and does **not** count. Budget-framed

59 answers convert at transcription: their monthly ? ÷ the matches/month they cited

60 (if they cited none, ÷4); the computed equivalent counts if ??300 and the

61 conversion is written into the rollup row. Stated WTP ? even clean ? remains

62 **talk**: the confirming step before any PROCEED-triggered build is the fake-door /

63 paid-pilot test already pre-agreed in PMF S6 risk 2 and the C13 row.

64

65 **Weighting rule (behavior > words):** H7-GREEN and H2-GREEN are **behavioral**

66 signals and count double in your gut-read; H3/H5 answers are **talk** until a Phase-2

67 session or a real payment corroborates them. A visit with H5-GREEN but H7-RED is a

68 polite conversation, not a customer.

69

70 **Disconfirming-evidence quota:** every visit sheet must record at least one

71 verbatim **negative** quote (the thing they liked least / wouldn't pay for). If a

72 sheet has only positives, the visit gets re-done or its evidence discounted in the

73 rollup (**never the associate's pay** ? they're paid per completed honest visit);

74 per the playbook's ground rule 4, a reasoned "no" is worth more than five vague yeses.

75

76 **Sampling note (read the verdict with this in mind):** the playbook sends the

77 associate to 10?15 **good** academies ? the verdict therefore reads "does the

78 beachhead work where it should work best?", not "what fraction of the market is
79 addressable". A PROCEED here licenses the next validation step, not a TAM claim;
80 the rollup's area/courts columns record what was actually sampled.
81

82 ## 3. Pre-registered decision rules
83

84 **Stopping rule (fixes the optional-stopping loophole):** the campaign targets
85 **n = 15** completed owner/head-coach sheets and may end earlier **only for
86 logistics** (associate availability, academies exhausted) ? **never because of how
87 the tallies look**. The 5-sheet check-ins below are progress reports, not verdict
88 opportunities; **the verdict is computed exactly once, at the final n**, and the
89 rollup records what that n was and why the campaign stopped. The count thresholds
90 below are written for n = 15 and apply unchanged at any final n in 10?14 ? which
91 deliberately makes the ?-bars *relatively harder* at smaller n (biases away from a
92 premature PROCEED) and the <-bars relatively harder to trip at n = 15 (biases away
93 from a premature FALSIFIED). Both biases err toward "keep investigating," the safe
94 failure mode. Counts, not percentages ? at this sample size we're reading
95 direction, not estimating proportions. "GREEN" below means GREEN on that
96 hypothesis row.
97

Verdict	Trigger (pre-registered)	What we then do
PROCEED	beachhead validated	?6 H2-GREEN **and** ?4 H5 at ?300+ (per the \$2 counting rule ? anchored numbers don't count) **and** ?3 H7-GREEN Run the camera pilot at the 2?3 warmest academies; those contacts seed the UI-alpha Wave-1 invite list (UI_ALPHA_EXECUTION_PLAN Phase R); the **fake-door/paid-pilot WTP test** (PMF \$6 risk 2) runs before money is spent on build
PIVOT A	"index the backlog" ? "record + index" service	<3 H2-GREEN **and** ?6 visits with H3 GREEN-or-AMBER **and** ?3 H5 at ?300+ The pain isn't the archive, it's that nothing useful gets made at all ? lead with the Phase-2 recording offer; camera pilot becomes the product wedge, not the demo
PIVOT B	partner/API motion	H4 RED at ?5 academies (incumbents entrenched and liked) Don't fight installed hardware head-on; explore the engine/API route through whoever owns the academy relationship ¹
PIVOT C	payer is the parent, not the academy	?4 visits where H5 is RED **and** the Q3 answer names parents/players as the watcher Re-test WTP as B2C per-student reels (?reel to parents) with the academy as channel, not customer
FALSIFIED	beachhead assumption fails	<3 academies record at all (H1) **and** 0 H7-GREEN The "academies already record fixed-angle" premise (PMF \$5 beachhead #1; the academy arm of \$6 risk 1) is wrong for this market ? revisit beachheads #2/#3 (pickleball-US, badminton Asia/Denmark) before building further

105

106 ¹ The template (surfaced by the 2026-06-10 competitive audit, not yet in PMF \$4's
107 web-sourced table): **PB Vision** sells pickleball video analysis B2C at ~\$99/yr
108 *and* as an ~\$8/hr-of-video partner API through facility platforms like
109 **PodPlay** ? the engine is ours, the academy relationship belongs to whoever
110 already owns it.
111

112 Mixed outcomes that match no row: write up the per-hypothesis tallies and decide
113 explicitly with the owner ? but **the tallies above get computed first**, so the
114 discussion starts from the pre-registered read, not from anecdotes.
115

116 **Competitor-intel side-channel (always-on):** any visit where a competitor is
117 named ? fill the answer sheet's competitor block (what it does, what it misses,
118 what they pay). ?3 consistent reports about one product = update
119 `PMF\_MARKET\_RESEARCH.md` \$4 with field-verified rows (today's table is web-sourced).
120

121 ## 4. Mechanics
122

- 123 - One [`FIELD\_VISIT\_ANSWER\_SHEET.md`](FIELD_VISIT_ANSWER_SHEET.md) per academy,
124 same day as the visit, in the associate's words + verbatim quotes.
- 125 - The owner (or Claude session) transcribes each sheet into one row of the rollup
126 table below (keep it in this doc or a sheet next to it); tallies refresh per 5
127 visits as **progress checks only** ? the verdict is computed once, at the final
128 n, per the \$3 stopping rule.

```

129 - Demo asset: the associate carries the real compiled reel
130   (`output/MahadevpuraSingles_highlights.mp4`, trimmed to ~60s) saved offline on
131   their phone, shown **only at the scripted moment (after Q6, before Q7)** ? *show,
132   never promise*. Demo reactions and improvement asks are **product feedback**, not
133   scored signals: they feed the build backlog verbatim, while the H1?H7 verdict
134   machinery stays anchored to the pre-demo answers (that's why the demo comes last;
135   the sheet records whether it was shown). Showing it right before Q7 also makes
136   the Phase-2 ask a reaction to something seen, not imagined ? a stronger
137   behavioral read.
138 - Student answers (sheet's student block) likewise feed product feedback and the
139   H3 (audience) gut-read; they don't enter the verdict tallies, and no student
140   contact details are ever collected.
141
142   ### Rollup table (one row per completed visit)
143
144   | # | Academy | Area | Courts | Role spoken to | H1 | H2 | H3 | H4 | H5 (?) | H6 | H7 |
Gut | Phase-2? |
145   |---|---|---|---|---|---|---|---|---|---|---|---|
146   | 1 | | | | | | | | | | | |
147
148   ### Tally ? decision read-out (progress check per 5 visits; verdict ONCE at final n)
149
150   | Metric | Count | Threshold in play |
151   |---|---|---|
152   | H2 GREEN (dead footage) | /n | ?6 ? PROCEED leg; <3 ? PIVOT A leg |
153   | H5 at ?300+ (pre-ladder / converted only ? $2 counting rule) | /n | ?4 ? PROCEED leg;
?3 ? PIVOT A leg |
154   | H7 GREEN (Phase-2 yes + contact) | /n | ?3 ? PROCEED leg; 0 ? FALSIFIED leg |
155   | H3 GREEN-or-AMBER (audience exists) | /n | ?6 ? PIVOT A leg |
156   | H4 RED (incumbent satisfied) | /n | ?5 ? PIVOT B |
157   | H1 any recording | /n | <3 ? FALSIFIED leg |
158   | H5 RED **and** Q3 names parents/players | /n | ?4 ? PIVOT C |
159
160   ## 5. Guardrails that bind the field work
161
162   - **Phase 1 records nothing** ? notes only (playbook rule 7). Student
163     conversations: coach-permitted **adult (18+) students only** ? "senior batch"
164     minors don't qualify; skip and note it when no adults are available.
165   - **Phase 2 consent rules are hard:** written consent from the academy AND adult
166     players; **sessions with minors are never recorded for our library** (training-pool
167     guardrail, CLAUDE.md); the academy is told plainly what the footage is for.
168   - Answer sheets contain personal contact details ? they stay between the associate
169     and the owner; transcribe to the rollup without phone numbers.
170

```

8. assistant (2026-06-21T03:17:29.581Z)

Now let me read the remaining docs in parallel.

9. user (2026-06-21T03:17:31.476Z)

```

1   # Field & Product Outreach Associate ? KhelSutra (Sports-Tech, Bengaluru)
2
3   **Help a Bengaluru sports-AI product meet its first real users.**
4
5   | | |
6   |---|---|
7   | **Duration** | ~3?4 weeks, part-time (flexible; mostly afternoons/evenings when
academies are active) |
8   | **Location** | Bengaluru ? in-person visits to badminton academies |
9   | **Pay** | ? ___ per completed academy visit + travel reimbursement; recording sessions
(stretch goal below) paid extra ? ? ___ per session |
10  | **Reports to** | Avi (phone/WhatsApp: ___ ) |
11
12  ---

```


The product, in plain words

KhelSutra turns ordinary practice/match video from a badminton academy into ready-to-use pieces: it finds the rallies in the recording automatically, cuts **highlight reels**, and produces a **simple summary** (how many rallies, how much actual play time). It works on video from a phone on a tripod ? no special cameras, no hardware to install. We're building a private web app that a few invited academies will be able to try.

The job, in one paragraph

Visit good badminton academies across Bengaluru and have honest conversations with **coaches AND students**: how do they work with video today, would something like this actually help them, and **what would make it genuinely useful for them** ? including what's missing or wrong with the idea. You work from a provided conversation guide, write down what people **actually say** (the negative answers matter most), and debrief with Avi after each visit. **You are gathering feedback, not selling.** No tech background needed.

The stretch goal (extra pay)

For academies that are interested, you coordinate a simple recording session ? a phone on a tripod, one-page checklist provided. We run that video through KhelSutra, and you go back and show them **their own court, their own players, as highlight reels and a report**: **"this is what it looks like for you."** Then you write down exactly what they say. There is no stronger feedback than that moment.

What you'll actually do

- Visit **10-15 academies** (intro + conversation, ~20-30 minutes each).
- Talk to the **owner/head coach** first, and ? with their permission ? a few **adult (18+) students** about whether they'd watch/share their own rally clips and what they'd want from them.
- Fill **one structured answer sheet per visit** (we provide it) ? their words, not yours, including the most negative thing said.
- Show a short sample reel on your phone **only at the point the guide says to** (after the open questions ? it changes people's answers if shown early).
- Debrief with Avi after each visit (WhatsApp/call). **Your first two visits are done together with him**, so you're never improvising.
- **Stretch:** coordinate recording sessions and present the sample results back.

You're a good fit if

- You're comfortable walking into an academy and starting a conversation with a stranger ? and just as comfortable hearing "no, this is useless to us."
- You can capture what people said **accurately** ? quotes, not summaries.
- Kannada / Hindi / English, as the conversation needs (you choose the mix).
- You play or follow a racket sport (helps a lot; not required).
- No technical skills needed. Reliability matters more than anything else.

What you get

- Paid per completed honest visit ? **not** per positive answer. A well-documented "they hated it, here's why" is a fully paid, fully successful visit.
- Ground-floor view of a real sports-AI product finding its market; your field notes directly shape what gets built next.
- Flexible hours, short commitment, clear scope.

Ground rules (the short version ? full version provided on joining)

1. Honesty over positivity. Never oversell, never promise features, prices, or dates.
2. **No recording of any person without written consent** ? and sessions involving **minors are never recorded** for our use, period. In regular visits you record nothing at all: notes only.

78 3. People's contact details stay between you and Avi.
79
80 ## How to apply
81
82 Message Avi at ____ with **FIELD**, one line about a sport you follow, and one
83 line about a time you had to get honest feedback out of someone. Next step: a
84 15-minute phone call with Avi.
85

10. user (2026-06-21T03:17:31.503Z)

1 # KhelSutra ? Product Overview & Early-User Guide
2
3 *What it does today, how to get the best out of it, and where it's going.*
4 *For field associates and for invited alpha / beta academies, coaches, and players.*
5
6 ---
7
8 ## In one line
9
10 ****KhelSutra turns ordinary badminton video into useful pieces ? automatically.**** Put a
phone or GoPro
11 on a tripod behind the court, record a session, and KhelSutra automatically finds the
rallies, stitches
12 a ****highlight reel****, and gives you a ****simple summary**** (how many rallies, how much
real play time).
13 No special hardware, no editing, no manual scrubbing.
14
15 ---
16
17 ## The problem we're solving
18
19 Academies and serious players already record a lot of video ? and then almost never
watch it. Hours of
20 footage pile up on phones and memory cards because cutting it into useful clips by hand
is tedious and
21 nobody has the time. The good rallies, the moments worth reviewing, stay buried.
22
23 Most academy halls also run ****several courts at once****, so a single recording is a busy,
noisy scene
24 that ordinary video tools choke on. ****Pulling the real rallies out of busy, multi-court
footage is the**
25 **hard problem KhelSutra is built to crack**** ? it already works very well on clean,
single-court
26 recordings, and the multi-court case is exactly what we're sharpening now.
27
28 ---
29
30 ## Who it's for
31
32 - ****Coaching academies & coaches**** ? turn raw session footage into review-ready clips
and reels for
33 teaching, without an editor.
34 - ****Serious players**** ? get clips of your own rallies to review, keep, and share.
35 - Today's focus is ****badminton academies in Bengaluru**** ? that's who we're building with
first.
36
37 ---
38
39 ## What it does ****today****
40
41 This is what's working right now, in the private web app invited users can try:
42
43 - ****Upload a full session or match video**** (from a phone on a tripod, or a GoPro). Long
videos are fine.

44 - ****Automatic rally detection**** ? it finds the rallies and where they start and end, on its own.

45 - ****Highlight reel**** ? the rallies stitched into one MP4 you can watch, download, and share, with a small

46 ****Generated by Khelsutra.guru**** mark.

47 - ****A simple summary**** ? how many rallies were found and how much actual play time there was.

48 - ****Re-use & download**** ? process once, come back to it later, download the reel.

49

50 That's the core loop: ****record ? upload ? get back a reel + summary.**** No editing skill needed.

51

52 > Like any automatic tool, it can occasionally miss a real rally or include one that isn't ? telling us

53 > when that happens is genuinely useful feedback, not a sign it's broken (more on that below).

54

55 ---

56

57 **## How to use it (alpha)**

58

59 1. ****Your KhelSutra associate helps you get access**** to the private web app (it's invite-only; today the

60 app is in English).

61 2. ****Upload your session video.**** Large files upload in pieces and can take a while on Indian uplinks ?

62 that's normal.

63 3. ****It processes in the background.**** You can close the tab and come back later; you don't have to sit

64 and watch it.

65 4. ****When it's done****, you get the ****highlight reel + summary**** to watch and download.

66 5. ****Re-process anytime.**** (Re-running the same video replaces its earlier result rather than keeping both.)

67

68 ---

69

70 **## How to get the best out of it (capture matters)**

71

72 KhelSutra works best on a ****steady, full-court recording**** ? the same way a coach would set a camera to

73 review play. A few simple habits make a big difference to the results:

74

75 - ****Put the camera behind the court, slightly elevated****, with the ****whole court in frame****.

76 - ****Keep it still**** ? a fixed tripod beats a handheld or panning shot every time.

77 - ****Shoot landscape****, with reasonable lighting.

78 - ****Record the whole session**** and upload it as-is ? KhelSutra does the cutting.

79

80 Footage that fights these habits (shaky handheld, heavy zooming/panning, broadcast-style cuts, very dark

81 halls) is the ****hardest case**** and where results suffer most. When in doubt: fixed camera, full court,

82 don't touch it. *(Your field contact provides a one-page capture checklist at setup.)*

83

84 ---

85

86 **## The part that makes KhelSutra different: **it gets sharper from real academy footage****

87

88 This is the heart of the product, and what sets it apart from a generic "AI clipper":

89

90 ****The footage that real academies contribute is what makes KhelSutra better over time**** ? especially on

91 the busy multi-court halls that are the hard case. When you choose to share your footage, and when

92 coaches confirm or correct what it found, that growing, labelled library is what we use
to sharpen the
93 tool. It's designed to improve in new versions as that library grows, rather than stay
frozen ? which is
94 why early users aren't just customers, they're co-builders.
95
96 ****Your data stays yours ? and sharing is your choice.**** We will ****never**** use your video
to improve the
97 tool without your ****clear, explicit agreement**** ? it is opt-in, never automatic.
****Footage involving**
98 minors is never used to improve the tool and is never published.****** We never publish your
footage; you
99 decide what, if anything, you share.
100
101 ****How your video is handled (plainly):**** these are large videos that need real
processing, so your video
102 is handled securely on our infrastructure to produce your reel. We don't publish it, and
we don't use it
103 to improve the tool without your opt-in.
104
105 > ****Read this honestly:**** today the tool gives you rally detection + reels + summaries.
It does ****not****
106 > retrain itself as you use it, and there is no per-person tracking. The "gets sharper
over time" above is
107 > the model we're building toward, powered by contributed footage ? not a switch that's
already on.
108
109 ---
110
111 **## Where it's going ? toward an **AI assistant coach****
112
113 Today KhelSutra finds rallies and makes reels. The roadmap ? fed by the growing video
library above ?
114 turns it into a genuine training companion:
115
116 - ****Shot detection**** ? recognising smashes, drops, net shots, serves, and clears within
each rally.
117 - ****Deeper per-rally and per-match analysis**** ? patterns, tendencies, what's working and
what isn't.
118 - ****Per-match history**** ? track progress over weeks and months, not just one clip.
119 - ****Coaching insights**** ? an ****AI assistant coach**** that supports (never replaces) the
real coach,
120 surfacing what to work on next.
121 - ****More sports**** ? table tennis next, then other racket/court sports.
122 - ****Your language**** ? Kannada and Hindi, alongside English.
123
124 > ****Read this honestly:**** the items in this section are the ****direction****, not today's
buttons. What you
125 > can use ***today*** is rally detection + highlight reels + the play-time summary. The
deeper analysis arrives
126 > as the video library grows ? which is exactly why your footage and feedback matter so
much.
127
128 ---
129
130 **## What feedback helps us most (for alpha users)**
131
132 You don't need to be technical ? just tell us, in your words:
133
134 - ****When it missed a real rally****, or ****grabbed something that wasn't a rally.****
135 - Whether the ****reel is the right length and order**** for how you'd actually use it.
136 - ****What numbers or details you wish the summary showed.****
137 - Your real ****capture conditions**** ? how many courts were running, camera/phone,
lighting.
138 - Anything that would make this ****genuinely useful for your academy**** (or a reason it

wouldn't be).

139

140 *(Send feedback however is easiest ? your KhelSutra associate will help you with that.)*

141

142 ---

143

144 ## What we're focused on over the next ~4 weeks

145

146 1. **Growing the library of real academy footage** ? especially busy **multi-court**
halls (the hard,

147 valuable case). This is the single biggest lever on quality.

148 2. **Field conversations with academies** ? understanding how coaches and players
actually work with

149 video, and what would make this genuinely useful.

150 3. **Recording sessions and showing academies their own footage** as reels ? *"this is
what it looks*

151 *like for you."*

152 4. **Sharpening multi-court accuracy** and inviting a few more academies into the
private trial.

153

154 ---

155

156 ## What KhelSutra is **not** (so expectations stay honest)

157

158 - It's an **early, invite-only trial** ? useful, but still improving; you'll hit rough
edges.

159 - It is **not** a line-calling, scoring, or umpiring system, and **not** player tracking /
identification.

160 - **Shot detection and the deeper "AI coach" analytics are on the roadmap, not available
today.**

161 - It is **not** a replacement for a coach ? it's a tool that saves coaches time and adds
insight.

162 - **Pricing isn't set** ? we're still learning what's fair; today's invited use is about
getting it right.

163

164 ---

165

166 ## Quick answers to common questions

167

168 ****"How accurate is it?"****

169 Very good on a clean, single-court recording. A busy multi-court hall is harder ? and is
exactly what

170 we're improving. The best answer is to *show* a sample reel rather than quote a number.

171

172 ****"Do I need a special camera or hardware?"****

173 No. A phone or GoPro on a tripod, behind the court, whole court in frame. We provide a
one-page checklist.

174

175 ****"Is my video private? What happens to it?"****

176 We treat your footage as private: we never publish it, and we never use it to improve
the tool without

177 your explicit opt-in. These are large videos that need real processing, so your video is
handled securely

178 on our infrastructure to make your reel. **Footage involving minors is never used to
improve the tool and**

179 **is never published.**

180

181 ****"Can it tell me the score / who won / call lines?"****

182 Not today. Today it finds rallies and builds a highlight reel + a play-time summary.

183

184 ****"Does it work for doubles and tournaments?"****

185 It finds rallies in doubles too. Tournament footage varies: a clean, fixed, full-court
recording works

186 well; crowded multi-court tournament halls are the hard case we're specifically working
on.

```

187
188     ***"Will it get better?"**
189     Yes ? that's the whole idea. The more real footage runs through it (with permission) and
the more coaches
190     correct it, the sharper future versions get, and the sooner shot detection and deeper
analysis arrive.
191
192     ***"Which sports?"**
193     Badminton today; table tennis and other racket sports are on the roadmap.
194
195     ***"When can I use it / how do I join?"**
196     It's an invite-only private trial. If you're interested, the best next step is to let us
record a session
197     and show you your own court and players as a reel.
198
199     ---
200
201     *KhelSutra · khelsutra.guru · Internal & invited-trial overview, June 2026.*
202     *Field associates: the role description is in the Field Associate flyer, and the
conversation script and
203     visit rules are in the separate Field Visit Playbook.*
204

```

11. user (2026-06-21T03:17:34.220Z)

```

1      # Field Visit Playbook ? academy conversations (KhelSutra)
2
3      > **Recruiting?** Hand candidates
[`PRODUCT_FIELD_ASSOCIATE_FLYER.md`](PRODUCT_FIELD_ASSOCIATE_FLYER.md)
4      > ? that's the role description. **This document is the working playbook the hired
5      > associate gets on day one:** the conversation script, the ground rules, and the
6      > Phase-2 (recording / show-back) procedure.
7
8      **Duration:** ~3?4 weeks, part-time (flexible hours, mostly afternoons/evenings when
academies are active)
9      **Location:** Bengaluru ? in-person visits to badminton academies
10     **Compensation:** ? ___ per completed academy visit + travel reimbursement; Phase-2
sessions paid extra ? ? ___ per session *(rates filled at hire; same numbers as the recruiting
flyer)*
11     **Reports to:** Avi (phone/WhatsApp: ___ )
12
13     ---
14
15     ## The gig in one paragraph
16
17     KhelSutra takes ordinary practice/match video from a badminton academy and finds the
rallies automatically, builds
18     highlight reels, and generates a simple summary (rallies found, actual play time); we're
building a private web app
19     that a few invited academies will be able to try. We need honest, on-the-ground feedback
from real academies ? **how
20     they work with video, whether this actually helps them, and what would make it genuinely
useful**. **Your job: visit
21     10?15 good academies, have a friendly 20?30 minute conversation with the owner/head
coach (and, with permission, a
22     few adult (18+) students) using the script below, and write down what they actually
say.** No selling. No tech knowledge needed. For interested academies there's a Phase-2 follow-
up
23     (a recording session ? then you return and show them THEIR OWN footage as highlights)
with additional compensation.
24
25     ---
26
27     ## Phase 1 ? The conversation (what you actually do)
28

```

29 **Who to talk to:** the academy owner or head coach ? the person who decides what the academy spends money on.

30 If you get a junior coach, be friendly, gather what you can, and ask when the owner is available.

31

32 **Opening (keep it this honest):**

33 > "Hi, I'm helping a Bengaluru engineer who's building an AI tool for badminton academies ? it takes normal practice

34 > video and automatically cuts it into rallies and highlights. He asked me to talk to a few serious academies to

35 > understand how you work with video today. Could I ask you a few questions? About 15?20 minutes, and we're not

36 > selling anything today."

37

38 **The questions (ask in this order; the starred ones matter most):**

39

40 1. Do you ever video-record sessions or matches here? How ? phone, tripod, CCTV, a hired person?

41 2. ? **Do you have recorded footage sitting around that nobody has time to watch?** (multi-court hall recordings,

42 tournament days, CCTV) ? roughly how much?

43 3. Who would watch rally clips or highlights if they existed ? coaches for training? players? parents?

44 4. What do you use today for any of this? (Listen for: Machaxi, VISIST, Game Theory, Stupa, BadPro+, just

45 YouTube/phone.) What do you like / what's missing in it?

46 5. ? **If you could hand over a full session's video and get back the rallies cut out + a highlight reel + a simple

47 summary ? would you pay per match for that? What would that be worth to you, per match?** Then **stop talking and

48 wait** ? let THEM name the number. If they genuinely won't after a pause and one "roughly?", you may offer the

49 ladder: "?100, ?300, ?500?" **On the sheet, tick whether their number came BEFORE or AFTER you offered the

50 ladder** ? it changes what the answer means. (Note their face, not just the words.)

51 6. Your hall has multiple courts running at once, right? Does that make video less useful today?

52

53 **? NOW (and only now) show the sample reel** ? a ~60-second real highlight reel we put on your phone. Say: "this is

54 what it produces ? the rallies found and cut automatically." Then shut up and watch their face. Write down their

55 FIRST reaction verbatim, and ask one follow-up: "**what would this need to do for it to be useful for YOUR academy?"**

56 (Improvement asks, in their words, go on the sheet. Showing it earlier anchors every answer above ? never lead with it.)

57

58 7. Would you let us record one session here (we pay a small fee for the time) and come back showing what the tool

59 produces on YOUR court, your players? *(This is the Phase-2 door ? just note yes/no/maybe and who to call.)*

60

61 **Students (with the coach's permission ? adult (18+) students ONLY, 5 minutes):** ask the coach to point out which

62 students are 18 or older ? "senior batch" players who are minors do NOT count. Then ask 2?3 of them:

63 (a) **Would you watch clips of your own rallies if they existed after each session?**

64 (b) **Would you share them ?

65 with whom?** (c) **What would you want them to show?** Capture their answers in the sheet's student block. Don't

66 collect students' contact details; if no adult students are available, skip the student questions ? note that instead.

67

67 **After each visit, fill ONE answer sheet ? the template is

[`FIELD\_VISIT\_ANSWER\_SHEET.md`](FIELD_VISIT_ANSWER_SHEET.md)

68 (print one per visit):** academy name, location, # courts, person spoken to + role +
 contact, answers 1?7 (their
 69 words, not yours), the most negative thing they said (required), exact quotes, your gut
 read, and whether they're a
 70 Phase-2 candidate. *(How the answers get scored and what totals trigger which decision
 is our job, not yours ?
 71 just capture honestly.)*
 72
 73 ---
 74
 75 ## Ground rules & gotchas (read twice ? this is the difference between useful and
 useless)
 76
 77 1. **You are researching, not selling.** If they ask "how much will it cost", say
 "that's exactly what he's trying to
 78 figure out ? what would feel fair to you?" Never quote a price yourself.
 79 2. **Past behavior beats opinions.** "Would you use video?" gets polite lies. "When did
 you LAST record a session, and
 80 what happened to that file?" gets the truth. Prefer the second kind of question.
 81 3. **Don't promise anything.** No features, no dates, no free trials. You may SHOW the
 standard sample reel (it's
 82 real output ? at the scripted moment only, after Q6) and you may offer the Phase-2
 idea (a paid recording session
 83 + free sample output of their own court), and only as "would you be open to it".
 Nothing else.
 84 4. **Write down disappointing answers verbatim.** A "no, we'd never pay for that" with a
 reason is worth MORE to us
 85 than five vague yeses. You're paid per completed honest visit, not per positive
 answer.
 86 5. **If they already use a competitor (Machaxi/VISIST/Game Theory etc.) ? goldmine.**
 Ask what it does well, what it
 87 doesn't, and what they pay. Spend extra time here.
 88 6. **Multi-court is our favorite topic.** If they complain that video is useless because
 "all the courts are in frame
 89 and it's chaos" ? capture that complaint in their exact words. That's the problem we
 solve.
 90 7. **No recordings of people without clear permission.** In Phase 1 you record NOTHING ?
 notes only. Don't photograph
 91 the academy or players unless the owner offers.
 92 8. **Don't oversit.** 20 minutes, thank them, leave a good impression ? we may come back
 to record.
 93 9. **Confidentiality:** don't describe the tech beyond the opening line, don't name
 other academies' answers, and
 94 don't share the answer sheets with anyone except Avi.
 95
 96 ---
 97
 98 ## Phase 2 (only for academies that said yes ? separate briefing & pay)
 99
 100 You coordinate a recording session: we pay the academy a small fee, set up a
 phone/camera on a tripod (we provide a
 101 one-page capture checklist: behind the court, elevated, whole court in frame, landscape,
 1080p/60fps, don't move it),
 102 record 1?2 sessions ? then we process the footage through KhelSutra and **you return to
 present the academy its own
 103 sample output** (their court, their players, as a highlight reel + summary, free) and
 fill a short reaction sheet:
 104 what they said when they saw it, what they'd change, whether they asked to keep using
 it. That show-back conversation
 105 is the single most valuable artifact this whole engagement produces. **If they ask "can
 we keep using it?"** ? that's
 106 the best possible signal: write it down verbatim, say **"I'll pass that straight to
 Avi"**, and promise nothing. **Hard rules:** written consent from the academy
 107 AND the adult players being recorded; **sessions with minors are not recorded for our
 library** ? adult sessions only;


```

108     the academy gets told plainly what the footage is for (building and demonstrating the
109     tool).
110     ---
111
112     ## Day one
113
114     Your first 2 visits are paired (you + Avi, or a live phone walkthrough) so you're never
115     improvising. Before visit 1:
116     read this playbook twice, have the sample reel saved offline on your phone (Avi provides
117     the ~60-second trim), and carry printed answer sheets
118     ([`FIELD_VISIT_ANSWER_SHEET.md`](FIELD_VISIT_ANSWER_SHEET.md) ? one per academy).
119

```

12. user (2026-06-21T03:17:34.643Z)

```

1     # Field Visit Answer Sheet ? one per academy (fill the same day)
2
3     > Print one per planned visit. Write THEIR words, not yours. A clear "no, because?"
4     > is a *good* sheet. Companion script: `FIELD_VISIT_PLAYBOOK.md`.
5
6     ## Visit basics
7
8     | | |
9     |---|---|
10    | Date / time | |
11    | Academy name | |
12    | Area (neighbourhood) | |
13    | # courts in the hall | |
14    | Spoke to (name + role) | |
15    | Contact (phone/WhatsApp) ? *only if offered* | |
16    | How long the conversation ran | |
17
18    ## The seven questions ? their answers (verbatim where possible)
19
20    **Q1 ? Do they record sessions/matches today? How (phone / tripod / CCTV / hired)?**
21
22    >
23
24    **Q2 ? ? Footage sitting around that nobody watches? Roughly how much, and from what?**
25
26    >
27
28    **Q3 ? Who would watch rally clips/highlights (coach / players / parents)? When?**
29
30    >
31
32    **Q4 ? What do they use today (Machaxi / VISIST / Game Theory / Stupa / BadPro+ / none)?
33    What's good, what's missing, what does it cost them?**
34
35    >
36
37    **Q5 ? ? Would they pay per match for rally cuts + reel + summary? What number did
38    THEY say (?)? What was their face doing?**
39
40    Their number came (tick one): ? BEFORE the ladder (they volunteered it) .
41    ? AFTER I offered ?100/300/500 . ? no number / refused .
42    ? they framed it differently (per student / per month ? write it below)
43
44    >
45
46    **Q6 ? Multiple courts in frame ? does that make video useless today? Their words:**
47
48    >
49

```

50 **Demo (shown AFTER Q6, never earlier) ? their FIRST reaction to the sample reel,
51 verbatim. Then: "what would this need to do to be useful for YOUR academy?" ?
52 their improvement asks, in their words:**
53
54 Demo shown? ? yes · ? no (why: _____)
55
56 >
57
58 **Q7 ? Open to a paid recording session on their court (we return showing THEIR
59 footage as highlights)? (yes / maybe / no ? and who to call)**
60
61 >
62
63 **Students (coach-permitted, adult 18+ ONLY ? skip and note it if no adult
64 students available): would they watch clips of their own rallies? share with
65 whom? what should clips show? Their words:**
66
67 >
68
69 ## Required: the most negative thing they said (verbatim)
70
71 >
72
73 ## Best quotes (anything vivid, in their words)
74
75 >
76
77 ## Scores ? OFFICE USE, leave blank (scored by Avi at transcription; when torn, score
lower)
78
79 | Hypothesis | Score |
80 |---|---|
81 | H1 recording behavior | GREEN / AMBER / RED |
82 | H2 dead footage | GREEN / AMBER / RED |
83 | H3 audience | GREEN / AMBER / RED |
84 | H4 whitespace | GREEN / AMBER / RED |
85 | H5 willingness to pay ? their ? number: _____ | GREEN / AMBER / RED |
86 | H6 multi-court pain | GREEN / AMBER / RED |
87 | H7 Phase-2 door | GREEN / AMBER / RED |
88
89 ## Competitor block (only if they named one)
90
91 | | |
92 |---|---|
93 | Product | |
94 | What it does well (their words) | |
95 | What it misses (their words) | |
96 | What they pay | |
97
98 ## Your gut read (circle): COLD · POLITE · CURIOUS · EAGER
99
100 ## Phase-2 candidate? (circle): YES ? call: _____ · MAYBE · NO
101
102 ## Anything else worth knowing (layout, lighting, camera positions you noticed, who
really decides)
103
104 >
105

13. assistant (2026-06-21T03:17:36.936Z)

Now let me read the remaining docs: PERSONALIZATION_PLAN, UI_ALPHA_EXECUTION_PLAN, and PMF_MARKET_RESEARCH.

14. user (2026-06-21T03:17:37.884Z)

```

1      # Personalization Plan ? a per-user feature on a generalization-first baseline
2
3      **Status: DESIGN ? owner-adopted (hybrid); Phase-0 engineering landing.** This is the
authoritative
4      plan for the personalization track. The detailed strategy analysis + the full pros/cons
that led here
5      live in the working artifacts `scratch/PERSONALIZATION_PIVOT_ANALYSIS.md` (the decision-
grade
6      counter-case) and `scratch/PERSONALIZATION_FEATURE_DESIGN.md` (the code-grounded
design); this doc is
7      the checked-in summary + decisions + phased build plan. Peer to
8      [OWNED_MODEL_IMPLEMENTATION_PLAN.md](OWNED_MODEL_IMPLEMENTATION_PLAN.md),
9      [EXPERIMENT_HARNESS.md](EXPERIMENT_HARNESS.md), and
[MULTI_SIGNAL_FUSION_PLAN.md](MULTI_SIGNAL_FUSION_PLAN.md).
10
11     ## 0. Thesis & strategic rationale (why *hybrid*, not personalization-first)
12
13     **Foundation principle (owner):** "a tool that gets better the more you help it."
Contribution
14     (data / video / diagnostics / annotation) improves BOTH the user's own model AND the
shared baseline;
15     **payoff scales with contribution.**
16
17     A directional pivot ? *personalization-first* (ship a baseline that continually self-
optimizes to one
18     user's own footage) ? was researched in depth. The verdict was **HYBRID, not all-in**:
19
20     - **Adopt personalization as a flag-gated FEATURE on a generalization-first baseline.**
The harness
21     genuinely makes per-user *tuning* near-free (it is already a corpus-relative A/B
engine; a per-user
22     corpus is just a special case). This is real and worth shipping.
23     - **Do NOT bet the strategy on per-user *learning*.** Per-user A/B is the n?6 small-
sample regime
24     *worse* (sharding data when you should pool); the differentiated parts (a learned per-
user model
25     serving path, the correction?label write path, a user/tenant dimension) are largely
unbuilt; a
26     naively-overfit per-user model loses the cross-user data flywheel and has no shared
baseline to
27     regress against.
28     - **The owner's contribution-flywheel + free-tier-pooling closes the one real hole** the
counter-case
29     found: by having (free-tier) users contribute to the *shared* baseline, the
compounding cross-user
30     moat is preserved *alongside* personalization ? personalization **and** flywheel, not
either/or.
31
32     Net: a generalization-first **baseline** ("batteries"), a thin **per-user tuning** layer
gated by the
33     honest small-N statistics (with a hard floor + structural-guard protection), and a
**contribution
34     loop** that feeds both.
35
36     ## 1. Owner decisions (locked 2026-06-14)
37
38     1. **Identity** ? auth WILL land; *design for the north star (multi-user / cloud),
implement for
39     current use (single-user local)*. `user_id` nullable now (NULL = local install),
device/install
40     UUID for the alpha; real auth + a `users` table later.
41     2. **Per-user model storage** ? inline `classifier_json` in the (local) DB now, but
**behind a load
42     interface** so that post-auth it is fetched from the **cloud-hosted DB run by the
service** that

```

```

43     generates base models and orchestrates the product. The local store is one
implementation of that
44     interface.
45     3. **`min_n` is a PROMOTION-confidence floor, NOT a service/segmentation gate.** A
brand-new user with
46     one video is ALWAYS fully segmented by the baseline batteries ? never denied, never
"give me more
47     videos first." `min_n` only gates whether a *per-user* A/B is trusted enough to
OVERRIDE a baseline
48     cue default *for that user*; below it (or if the CI / sign test don't clear) ? keep
the baseline.
49     **Training-corpus policy:** *no more than needed, no less than all-if-few* ? fit on
the
50     `smallest_k_with_lift` videos (the held-out-USER learning curve), else ALL available.
51     4. **"Flip Gate-A to default" = ship the validated Gate-A held-shuttle guard as the
**default
52     detection path** (today it lives in the opt-in `FusionSegmenter`, which is not the
default
53     segmenter). Changes served behavior ? **gated on a golden-set regression run + owner
OK**; still
54     open (see §5).
55     5. **Flywheel = backend property** (not in-product marketing). A **free tier** is likely
and
56     **free-tier videos ARE pooled** into the baseline (consent baked into the free terms;
57     minors-exclusion still enforced) ? the clean value-exchange that feeds the owner-
trained base model.
58     Paid tiers get stronger privacy / opt-out.
59
60     **Pending sub-decision:** structural-guard scope ? recommended **Gate-A only** (the
proven structural
61     held-shuttle guard; its value is in failure modes a user's corpus may not yet contain),
with **Gate-B**
62     (person-occupancy / `min_players`) left *subject* to per-user A/B; **`min_n` = 5`**.
**Gate-A-only
63     structural, `min_n` = 5 ? owner-confirmed 2026-06-14.**
64
65     ## 2. Architecture
66
67     **Two-layer decoupling (already real).** Detection (GPU, run once): `FusionSegmenter`
persists the
68     **full superset** of per-window features into `candidate_segments.stats`; only the
served handoff is
69     gated. Decision (CPU, swappable): `experiment.Variant` + `predict()` re-score persisted
features.
70     **Personalization == a per-user `Variant`** (cue_flags + `min_crossings`/`min_players` +
optional
71     LogReg + threshold). No segmenter rewrite ? a per-user `FusionConfig` overlay resolved
upstream.
72
73     **The single highest-leverage wiring:** `fusion_hybrid.py::_select_for_handoff` is the
production
74     decision seam. When a per-user model is pinned AND the promotion gate says ON, apply
75     `LogisticRegression.predict_proba` over `window_stats` (the same math
`experiment.predict()` runs
76     offline). The **load half of the bridge already exists** ?
77     `backend/pipeline/personalization/serving.py::load_user_classifier` rehydrates the
stored
78     `classifier_json` (and `resolve_user_fusion_config` overlays the per-user
`fusion_config`) from
79     the v5 store ? but it is **not yet wired into the served path**: `_select_for_handoff`
still reads
80     only the static `idx_cfg["fusion"]`. Closing that one seam (the first change to served
behavior, so
81     owner-gated) is the remaining #1 wall.
82

```

```

83     **Per-user data model** (migrations follow the `PRAGMA user_version` ladder; current =
v5):
84     - **v4** ? ? **shipped.** Nullable `user_id` on `videos` + `candidate_segments` (NULL =
local;
85         byte-identical to today).
86     - **v5** ? ? **shipped.** `user_models` store: `(user_id, version, baseline_version,
87         feature_schema_hash, fusion_config, classifier_json, promotion_evidence,
n_videos_at_fit,
88         is_active)`. Inline `classifier_json` (tenant-safe); `feature_schema_hash` is the
89         MINOR-vs-MAJOR upgrade switch.
90     - **v6** ? **pending.** `contribution_consent` ledger (default-deny; minors-exclusion
enforced **at
91         the pooling query**, not advisory).
92
93     ## 3. Constructs (the gap list + what is built)
94
95     | # | Construct | Status |
96     |---|---|---|
97     | 1 | Per-user store + serving bridge ? **store + load half built** (v5 `user_models` +
`serving.py::load_user_classifier`); only wiring `_select_for_handoff` remains | Phase 1 (the #1
wall ? load half done, wiring pending) |
98     | 2 | Correction-loop ? per-user labels (`human_label` setter + confirm/reject API;
recall-protecting de-bias prior) | Phase 2 (`human_label` is a read-only stub today) |
99     | 3 | **Per-user promotion gate** (min-N floor + structural-guard exemption) | ? **DONE
? PR #141** (`backend/eval/per_user_gate.py`) |
100    | 4 | Contribution flywheel (opt-in data/video/diagnostics ? shared baseline; payoff ?
contribution; consent + minors-exclusion) | Phase 2 |
101    | 5 | User-as-annotator charter (rally-annotator: auto-generate a ~2-min high-
uncertainty segment from the user's own video; guide labeling ? grounding) | Phase 3 |
102    | 6a | **Held-out-USER learning curve** (the falsifiable "<K videos to superior quality"
criterion) | ? **DONE ? PR #142** (`backend/eval/per_user_eval.py`) |
103    | 6b | Per-user no-regression guard (frozen per-user CONTROL; never accept a regressing
retrain/upgrade) | ? **in flight** (`feat/per-user-no-regression`) |
104    | 7 | Day-0 "batteries" = the validated label-free Gate-A/B fusion (?F1 +0.074, 6/6,
p=0.031) shipped as the default | Phase 0, owner-gated ($5) |
105
106    ## 4. Phased PR-sized build plan + status
107
108    **Phase 0 ? safety + day-0 batteries** (no schema / API / served-behavior change; pure-
Python + tests):
109    - PR-0.1 per-user promotion gate ? ? **#141 merged**
110    - PR-0.2 held-out-USER learning curve ? ? **#142 merged**
111    - PR-0.3 per-user no-regression guard ? ? in flight
112    - PR-0.4 ship Gate-A as the day-0 battery ? **owner-gated** ($5; needs a golden-set
regression run)
113    - PR-0.5 within-user LOVO measurement (the "confirm-or-kill" experiment) ? **owner-
gated** (needs multi-video-per-user data on the GPU box)
114
115    **Phase 1 ? per-user tuning serving + store:** ? already landed ? v4 migration (nullable
`user_id`),
116    v5 `user_models` + CRUD (`upsert_user_model` / `get_active_user_model`), and
`serving.py`
117    (`resolve_user_fusion_config` overlay resolver + `load_user_classifier`). **Remaining:**
`set_user_id`
118    (and the Phase-2 `set_human_label` writer); **wire the bridge into `_select_for_handoff`
(owner-gated
119    ? served path)**; `/retune` + `/model` trust-feature endpoints.
120
121    **Phase 2 ? correction loop + flywheel:** `POST /api/candidates/{id}/label` (the first
`human_label`
122    writer); recall-protecting de-bias prior; v6 consent ledger + minors-exclusion; **opt-in
contribution
123    upload (owner-gated ? privacy surface)**; free-tier pooling.
124
125    **Phase 3 ? annotator charter + taxonomy + baseline upgrades:** active-learning task

```

```

picker; rally-
126     annotator CSV ingest ? per-user labels; **coarse hard/soft shot proxy (owner-gated)**;
**baseline-
127     upgrade governance (MINOR carry / MAJOR re-fit on `feature_schema_hash`) + optional
federated nudge
128     (owner-gated, counsel).**
129
130     ## 5. Open decisions (owner)
131
132     1. **Structural-guard scope + `min_n` value** ? lock **Gate-A-only structural, `min_n =
5`** (recommended)?
133     2. **Gate-A default flip (PR-0.4)** ? Gate-A lives in the non-default `FusionSegmenter`,
so "ship as
134         default" is a path choice (make that segmenter the default vs port the gate into the
default path).
135         Recommended sequence: **run a golden-set regression first**, bring the number, *then*
approve.
136     3. **Per-user `user_id` for the alpha** ? device/install UUID acceptable, or must auth
land first?
137     4. **Local-install flywheel default** ? fully-local zero-pooling until consent is
explicitly granted?
138
139     ## 6. Honest risks (carried from the analysis)
140
141     - **Small-N per-user eval noise** is central and *worse* per-user ? mitigated
structurally: the gate
142         refuses below `min_n` and falls back to the baseline; never auto-promotes in the
regime the tooling
143         exists to refuse in. PR-0.5 confirms-or-kills the core premise before any serving
build-out.
144     - **Cold-start** ? the day-0 Gate-A battery (zero per-user labels) is the floor for the
median user.
145     - **Privacy deferral** ? do not advertise "data never leaves your machine" until on-
device ships;
146         near-term pooling carries the full cloud (DPA / consent / minors) posture.
147     - **Compute** ? heavyweight VLM personalization (on-device Qwen3-VL fine-tune) is **out
of v1 scope**;
148         v1 is the cheap LogReg / threshold / cue-ON-OFF layer only.
149
150     ## 7. References
151     - `scratch/PERSONALIZATION_PIVOT_ANALYSIS.md` ? the decision-grade strategy analysis
(pros/cons, harness-fit, counter-case, recommendation).
152     - `scratch/PERSONALIZATION_FEATURE_DESIGN.md` ? the code-grounded design (schema,
serving bridge, APIs, critical files).
153     - Built: `backend/eval/per_user_gate.py` (#141), `backend/eval/per_user_eval.py` (#142).
154     - Harness: [EXPERIMENT_HARNESS.md](EXPERIMENT_HARNESS.md), `backend/eval/experiment.py`,
`backend/eval/eval_stats.py`.
155

```

15. user (2026-06-21T03:17:38.286Z)

```

1     # UI Alpha Execution Plan ? KhelSutra Guru
2
3     **Status:** ACTIVE · updated 2026-06-10 · **Linked:** `UI_PROPOSAL.md` ·
`HOSTING_PLAN.md` ·
4     `mockups/ui_alpha_mockup.html` · ADR-009 (`archives/decisions/DECISIONS.md`).
5     **Sequencing principle (owner-requested):** build and test **everything locally first**;
6     deployment (DNS/tunnel/Pages/Access) starts only after the local end-to-end gate passes.
7
8     ## Decisions locked
9
10    | Date | Decision |
11    |---|---|
12    | 2026-06-10 | Product name **KhelSutra Guru** · domain **khelsutra.guru** (GoDaddy,
3-yr + protection) |

```

```

13      | 2026-06-10 | Separate **private repo `khelsutra`** ("heart and brain"): `web/` SPA
now, control plane at beta; backend enablers stay in the indexer repo |
14      | 2026-06-10 | **Local-first delivery** (this plan's phase structure) |
15      | open | Final "go" on PR-1 coding start (= #16 greenlight) · training opt-in default
OFF · invite list · stack (React+Vite+TS+Tailwind stands unless owner objects) |
16
17      ## Master checklist
18
19      ### G0 ? Setup (now)
20      - [x] Name/domain/repo-split ratified; spelling confirmed via GoDaddy receipt
21      - [x] `khelsutra` repo **scaffolded** at `C:\Users\avidu\Projects\khelsutra` (README,
.gitignore, web/, docs/)
22      - [x] **Owner:** git-init + private GitHub push ? done 2026-06-10:
23        `https://github.com/avidullu/khelsutra` (private; git runs Windows-side per ADR-009)
24      - [x] **Owner:** explicit "go" for PR-1 ? given 2026-06-10
25      - [x] **Owner:** Gemini API key rotated ? 2026-06-10, owner-verified (confirm old-key
deletion in AI Studio)
26
27      ### Phase L ? build & test locally (no cloud; ?6?7 sessions)
28
29      **L1 ? backend enablers** (indexer repo · branch off master · ONE PR each, no stacking ·
offline tests in every PR · GPU/real-video checks = owner):
30      - [x] **PR-1 · B1 async job model (#16):** `jobs` table (user_version 1?), `POST
/api/process` ? 202 + `job_id`, `GET /api/jobs/{job_id}`, ThreadPoolExecutor worker,
restart/crash state handling ? **MERGED 2026-06-11, PR #87** (CI green: 333 passed; machine-side
real-video checks fold into L3)
31      - [x] **PR-2 · B2+B3 upload & download:** chunked upload (init ? sequential ?95 MB parts
? complete ? MD5; caps + mp4/mov allowlist; per-user dirs deferred to PR-3) + streaming `GET
/api/highlights/{video_id}/{name}` ? **MERGED 2026-06-11, PR #99** (also fixes static-UI
`segment_ids` compile bug + adds browser download link). **Live E2E verified at merge:** 86 MB
real video uploaded in 3 sequential parts and reassembled to MD5 byte-identity; out-of-order 409
/ traversal 400 (handler) + 404 (router) / unknown-id 404 confirmed; compile of a real confirmed
rally ? streamed ? download (3.9 MB reel).
32      - [ ] **PR-3 · B4+B7 identity & exposure:** `owner_email` column + scoping on every
query; Access-JWT/header read with `DEV_USER_EMAIL` env fallback for local dev; CORS narrowed;
rate limit; per-user quota
33      - [ ] **PR-4 · B6 correction loop:** `PATCH /api/segments/{id}` accept/reject/boundary-
nudge ? `candidate_segments.human_label`/`notes` + provenance (who/when); retire
`backend/api/static` (B5)
34
35      **L2 ? SPA** (khelsutra repo, `web/`; dev server proxies to `http://localhost:8000`):
36      - [ ] **M1 scaffold:** Vite + React + TS + Tailwind, router, typed API client, branding
(KhelSutra Guru)
37      - [ ] **M2:** A2 Upload (chunked, guidelines checklist) + A3 Job Progress (poll)
38      - [ ] **M3:** A1 Matches + A4 Rally Review (player seek, accept/reject/nudge wired to
PR-4)
39      - [ ] **M4:** A5 Highlights (compile + download) + A6 Report + engine-offline banner
40      - [ ] Unit tests for API client + review-state logic; `npm run build` clean
41
42      **L3 ? local end-to-end (owner's box, real match videos):**
43      - [ ] Full UI flow on localhost: upload ? job ? review/correct ? compile ? download
44      - [ ] `python run_all_tests.py` green; UI polling vs SQLite WAL sanity; WASB/CLI paths
untouched
45      - [ ] **GATE GL:** owner signs off local E2E ? only then deployment work begins
46
47      ### Phase D ? deployment (?2 sessions + ~1 hr owner)
48      - [ ] **D1 (owner):** Cloudflare account ? add khelsutra.guru ? GoDaddy nameserver
switch (HOSTING_PLAN $5) ? scoped API token ($6) ? Pages project connected to GitHub `khelsutra`
(build root `web/`)
49      - [ ] **D2 (Claude):** DNS records; `cloudflared` Windows service on the box ?
`api.khelsutra.guru`; Access apps + email-allowlist policies for `app` & `api`; SPA prod env
50      - [ ] **D3:** deployed smoke E2E (owner-only allowlist); go-live security checklist
(HOSTING_PLAN $7) all green
51      - [ ] **GATE GD:** checklist signed off
52

```

```

53     ### Phase R ? rollout
54     - [ ] Wave 1: 5 testers (C13/camera-pilot contacts) + feedback channel (WhatsApp group
or Google Form)
55     - [ ] 2-week retro ? punch list ? widen toward 25 ? beta planning (UI_PROPOSAL §4)
56     - Metrics: matches/week · % rallies corrected · median turnaround · verdict mix · week-2
tester retention
57
58     ## Working agreement
59     Claude (Cowork) designs, codes, writes tests + docs, and leaves a
`scratch/PRn_HANDOFF.md` per
60     work item. **Claude Code executes all git/PR mechanics** (branch, commit, push, PR,
conflict
61     fixes, merges after owner review) by reading that handoff ? the owner never copy-pastes
62     commands (ratified 2026-06-11 after PR-1/#88 friction; supersedes the command-block
pattern).
63     Cowork's sandbox cannot run git on mounts (ADR-009) and CI runs the offline suite on
every PR.
64     One PR per item; no stacking; never push to master directly. PowerShell note: no `&&`.
65
66     ## Risks & mitigations
67     Box uptime (offline banner, alpha-acceptable) · Gemini 429 burst-throttle (serial
workers,
68     `ai_max_workers=1`, 6-attempt backoff ? learned 2026-06-10) · CF free-tier 100 MB body
cap
69     (chunked upload design) · single-GPU box (alpha default = Gemini path; WASB jobs queue-
depth 1) ·
70     .guru renewal cost ?18.8k/3-yr (optional: transfer to Cloudflare Registrar at-cost after
the
71     60-day registration lock; domain-protection may need toggling for transfer).
72
73     ## Session map (default order)
74     S1?S2 PR-1 · S3 PR-2 · S4 PR-3 · S5 PR-4 · S6 M1+M2 · S7 M3 · S8 M4 · S9 L3 E2E + fixes
·
75     S10 D1?D3 · S11 wave-1 launch
76

```

16. assistant (2026-06-21T03:17:41.881Z)

Now let me read the PMF_MARKET_RESEARCH.md and the I18N_PLAN.md (referenced as the localization companion) to round out the buyer/user and language picture.

17. user (2026-06-21T03:17:42.852Z)

```

1     # PMF & Market Study ? AI Highlight-Reel Tool for Self-Recorded Racket Sports
2
3     > **Research snapshot ? 2026-06-07.** Figures are cited as-of with source + year;
4     > validate live before relying on them for investment/pricing decisions. This is the
5     > market-research-of-record behind the PMF section of the living
6     > [`COMMERCIALIZATION.md`](COMMERCIALIZATION.md) tracker (§6). Product framing:
7     > a closed-source hosted API/SaaS (plus local mode) that ingests long-form,
8     > self-recorded, **static-camera, full-court amateur** racket-sports video and
9     > auto-detects rallies and compiles highlight reels. Primary sport: badminton;
10    > sport-generic architecture roadmapped to tennis, pickleball, table tennis, cricket.
11
12    > **Confidence convention.** Figures from named industry reports, federations, or
13    > company pricing pages are tagged with source + year. Inferred items are tagged
14    > ***(estimate)** / ***(low confidence)***. Emerging-category market sizes vary widely
15    > between vendors; ranges are shown rather than false precision.
16
17    ---
18
19    ## 1. Executive summary
20
21    The product sits at the intersection of three favorable trends: (a) a fast-growing
22    **sports analytics market** (~USD 5?6B in 2025, ~18?22% CAGR to USD 24?36B by ~2034

```


depending on vendor); (b) a structural shift to **amateur self-recording** on action cameras and phones (action-camera market ~USD 6.5B in 2024, sports the largest segment); and (c) **strong racket-sport participation** ? 220M+ badminton players, 106M tennis players (2024), and a near-vertical pickleball curve (~19.8M US players in 2024, +45.8% YoY).

The central tension is the **static-camera constraint**: the tool works best on fixed, full-court amateur footage and **fails** on broadcast/panning footage. This is both a moat (most competitors target team sports / broadcast) and a filter on the addressable user ? the customer must already set a camera on a tripod behind the court. That behavior is concentrated in **coaches/academies, clubs, and serious enthusiasts**, not the median casual player.

Recommended beachhead (strongest single): **badminton coaching academies and serious club/competitive players in India** ? a large, rapidly formalizing academy ecosystem (Khelo India tailwind), genuine fixed-angle recording behavior, existing willingness to pay for coaching, an underserved tooling gap (incumbents are team-sports/broadcast), and a direct-but-immature competitor field. The product is **native badminton-first** ? a genuine edge here. Secondary: **(2) US/global pickleball players & clubs** (capital-rich, fastest-growing, but SwingVision already strong); **(3) badminton academies/clubs in East/SE Asia + Denmark** (highest density).

Pricing clusters into hardware-bundled auto-filming (Veo, Trace, Pixellot, Spiideo) and software-only phone AI (SwingVision USD 179.99/yr; BadPro+ pay-per-use). A software-only BYO-camera model undercuts the hardware players; realistic anchors **(estimate): B2C USD 60?180/yr; B2B academy/club USD 300?1,500/yr per location.**

Top risks: (1) amateurs won't reliably set up a tripod ? TAM collapses to coaches/clubs; (2) free/near-free alternatives cap B2C WTP; (3) capture-quality variance is the core failure mode; (4) badminton's strongest markets (India, Indonesia, China) are the most price-sensitive and hardest to bill at Western SaaS prices.

2. Market sizing & trends

2.1 Sports analytics / sports-video market

- 2025 size:** ~USD 5.47B (Precedence Research) to ~USD 5.79B (Fortune Business Insights); other vendors USD 5.6?6.0B. **(2025)***
- Growth:** CAGR 15.7%?22.5%. Precedence ? ~USD 29.75B by 2034 (20.63%); Fortune BI ? USD 24.03B by 2032 (22.5%); MarketsandMarkets (narrower) USD 4.75B by 2030 (15.7%); Future Market Insights ? ~USD 36.2B by 2035 (22.1%). **(reports 2024?2025)***
- Video analytics** is the leading segment within sports analytics (Grand View / Fortune BI) ? the directly relevant sub-market.
- Interpretation (estimate):** the **amateur/club auto-highlight** slice is a small but fast-growing fraction; most analytics revenue today is pro/broadcast/team-sports. The product targets the long tail the big numbers under-serve.

2.2 Action-camera / amateur recording adoption

- 2024 size:** ~USD 6.5B (GMIInsights) ? ~USD 16.9B by 2034 (~10% CAGR). **(2024?2025)***
- Sports** is the largest end-use segment of action cameras (GMIInsights, 2024).
- GoPro ~47% share, DJI ~20%** (electroIQ/industry, 2024); phones + cheap tripods are the de-facto amateur rig.
- Implication:** the capture-side behavior the product depends on is already widespread and growing ? **for the segments that record** ? validating "BYO-camera, software-only."

2.3 Participation & popularity by sport and geography

Sport	Participation signal	Source (year)
Badminton	220M+ play regularly; 160+ countries BWF est. via BadmintonBites (2023)	
Badminton ? China	~80M registered, up to ~100M (largest base) industry compilations (2023)	

86 | Badminton ? India | 20M+ recreational by 2023; academy boom + Khelo India | industry
compilations (2023) |

87 | Badminton ? Indonesia | National sport; ~1M club players | industry (est.) |

88 | Badminton ? Denmark | 100,000+ registered club members (Europe #1) | industry (2023) |

89 | Badminton ? Japan/Korea | Major school-club + workplace sport | industry (2023) |

90 | **Tennis** | **106M players (2024)**; +25.6% over 5 yrs (87M in 2021) | ITF Global
Tennis Report (2021, 2024) |

91 | Tennis ? Asia | ~1/3 of players (33M of 87M) | ITF (2021) |

92 | **Pickleball ? US** | **19.8M players (2024)**; +45.8% YoY, +311% over 3 yrs; ~6.2M
core | SFIA / Pickleheads / The Dink (2024) |

93 | Pickleball ? US courts | ~70,000 courts, +55% places to play YoY | SFIA / Pickleheads
(2024) |

94 | Pickleball ? global | IFP 78 member countries; UK +73% in 2024; ~282M played monthly
per PPA claim **(low confidence)** | various (2024?2025) |

95 | **Table tennis** | 30M+ players; 227 member associations | ITTF (current) |

96 | **Cricket** | 2.5B+ fans; 500M+ in India (mostly *fans*, not self-recorders) | Cricket
Rise / Nielsen (2024?2025) |

97

98 - **Pickleball market value:** ~USD 2.2B (2024) ? USD 9.1B by 2034 at 15.3% CAGR
99 (Market.us/Custom Market Insights, 2024?2025).

100 - **Badminton equipment market:** ~USD 7.23B global (2023), Asia-Pacific dominant.
101

102 **Geographic read:** badminton density is overwhelmingly Asian + Denmark; tennis is
103 broad/high-income; pickleball is US-dominant, capital-rich, globalizing fast; cricket
104 is huge as *fandom* but its amateur self-recording + static-full-court fit is weak
105 (large field, not a fixed-frame court). For *this product's static-camera fit*,
106 badminton/pickleball/tennis/table-tennis are the core; cricket is a long-tail roadmap
107 item, not a beachhead.
108

109 ---
110

111 ## 3. Customer segment evaluation
112

113 | Segment | Market-size signal | Willingness to pay | Accessibility | Static-camera fit
| B2C/B2B | Verdict |

114 |---|---|---|---|---|---|

115 | **Individual amateur/enthusiast** | Largest raw TAM (100s of millions) | **Low** ?
competes with free; ~USD 5?15/mo ceiling | High via app stores; high CAC/churn | **Mixed** ?
most won't set a tripod | B2C | Volume play, weak economics; use as funnel |

116 | **Coaches & coaching academies** | India academy boom; thousands across Asia |
High ? already sell coaching; USD 300?1,500/yr plausible | Reachable via
federations/distributors/social | **Strong** ? already film fixed-angle | B2B/B2B2C | **Best
fit** ? buyer, recorder, payer aligned |

117 | **Clubs & leagues** | Denmark 100k+; thousands of clubs Asia/EU | Medium-high; USD
500?3,000/yr | Medium ? committee sales cycle | **Strong** | B2B/B2B2C | Strong secondary;
longer cycle |

118 | **Tournament/event organizers** | Many small/regional events | Medium ? per-
event/seasonal | Medium ? episodic | **Strong** | B2B | Good per-event upsell; lumpy |

119 | **Schools & universities** | Large (esp. Asia/Japan clubs) | Low-medium ? budget-
constrained | Low ? slow procurement | Strong | B2B | Slow; grant/Khelo-India dependent |

120 | **National/regional federations** | Few, high-value | Medium-high; procurement-
heavy/political | Low ? long cycles | Strong | B2B | Lighthouse/credibility, not beachhead |

121 | **Broadcasters / streamers** | High value per deal | High | Low ? incumbents
entrenched | **Poor ? panning = failure mode** | B2B | **Avoid** ? violates core constraint |
122

123 **Key dynamics:** the static-camera constraint is a **B2B filter** ? coaches/academies/
124 clubs already record from a fixed angle, so the product fits existing behavior (cleanest
125 PMF). B2C is a large low-WTP, high-CAC funnel best used to feed B2B. Broadcasters
126 explicitly violate the constraint ? exclude.
127

128 ---
129

130 ## 4. Competitive & pricing benchmarks
131

132 | Product | Category | Model | Price | Notes (year) |

```

133 |---|---|---|---|---|
134 | **Veo (Cam 3)** | Team-sports auto-filming | Hardware + sub | Camera ~USD 1,533 (5G
~1,886) + ~£395?895/yr | 15,000+ clubs/80 countries; $115M raised (2025) |
135 | **Trace** | Soccer auto-filming | **Free camera + sub** | USD 180/yr (Basic) ? 300
(Pro) | Camera leased free with sub (2025) |
136 | **Pixellot (Air)** | Auto camera/streaming | Hardware + sub | Camera ~USD 949 + USD
69?167/mo | Institutional/league; 10,000+ clubs (2025) |
137 | **Spiideo** | Auto-filming/analytics | Sub incl. camera | ~USD 3,800/yr (1 team) ?
6,100/yr (club) | Institutional budgets (2025) |
138 | **Hudl** | Team video analysis | SaaS tiers | From ~USD 400/yr | Broad team-sports
incumbent (2026) |
139 | **SwingVision** | **Tennis/pickleball AI, single iPhone** | Software freemium | **USD
179.99/yr**; free 2 hrs/mo | Closest analog; 200k+ monthly users; $10M+ raised (2025) |
140 | **BadPro+** | **Badminton AI on phone** | **Pay-per-use, no sub** | Per-analysis (? "<
one coaching session") | Direct badminton competitor; shot detection, heatmaps, auto-highlights
(2025) |
141 | Niche pickleball/badminton apps | Various | Mixed | Varies | Fragmented; few true
auto-highlight tools (2025) |
142
143 **What buyers actually pay (synthesis):** consumer software-only anchors at ~USD
144 100?200/yr (SwingVision 179.99); hardware-bundled USD 180?900/yr prosumer/club,
145 USD 3,800?6,100/yr institutional; badminton-specific BadPro+ uses *pay-per-use* ?
146 signaling a real badminton AI market but **unproven subscription WTP** (opportunity +
147 warning). **Positioning:** software-only + BYO-camera undercuts hardware players and
148 sells worldwide via SaaS. Realistic anchors **(estimate): B2C USD 60?180/yr; B2B
149 academy/club USD 300?1,500/yr** per location. The differentiators (cheap offline
150 self-improving detection, per-rally stats/tagging, clean licensing) support a **B2B**
151 value story more than a B2C one.
152
153 ---
154
155 ## 5. Recommended beachhead
156
157 Ranked by **WTP × participation density × existing recording behavior × static-camera
158 fit × competitive whitespace.**
159
160 ### #1 (strongest) ? Badminton coaching academies & serious players, **India**
161 20M+ recreational players (2023); rapidly **formalizing academy ecosystem** + Khelo
162 India tailwind; academies already record fixed-angle; coaches are buyers who monetize
163 instruction (high relative WTP); incumbents ignore badminton; the only direct badminton
164 AI (BadPro+) is early with unproven pay-per-use. Product is *native badminton-first*.
165 **Watch-outs:** price sensitivity (need INR-tuned tiers); sell to the **academy**, not
166 the individual.
167
168 ### #2 ? Serious pickleball players & clubs, **US (then UK/Australia/Asia)**
169 Fastest-growing sport (19.8M US, +45.8% YoY, 2024); capital-rich; ~70,000 courts;
170 strong recording/sharing culture; pickleball is on the roadmap. **Watch-out:**
171 **SwingVision already owns single-camera tennis+pickleball AI** at USD 179.99/yr (200k+
172 monthly users) ? head-to-head differentiation, not greenfield.
173
174 ### #3 ? Badminton academies/clubs, **East/SE Asia (China, Indonesia, Malaysia, Japan,
Korea) + Denmark**
175 Highest global density (China 80M+; Denmark 100k+ structured club members; Japan/Korea
176 school-club & workplace systems), mature club infra, fixed-court recording.
177 **Watch-outs:** China needs local hosting/partners + is regulatory-complex; SE Asia
178 price-sensitive; Denmark/EU small but high-WTP and easy to pilot.
179
180 **Bottom line:** lead with **India badminton academies** (whitespace + behavior + buyer
181 alignment), use **enthusiast B2C as the funnel**, validate a **club/academy B2B2C
182 motion**, then expand to **pickleball-US** (out-build SwingVision) and
183 **badminton-Asia/Denmark** (density).
184
185 ---
186
187 ## 6. Risks & open questions (what would invalidate the thesis)

```

188

189 1. ****Tripod/static-camera adoption among amateurs.**** If serious amateurs won't set up a
190 fixed camera, B2C TAM collapses to coaches/clubs. **Validate:** observe 325 academies +
191 50?100 enthusiasts on whether they already record fixed-angle.

192 2. ****Free / near-free alternatives**** (YouTube, free app tiers, BadPro+ pay-per-use) cap
193 B2C WTP. **Validate:** fake-door pricing at USD 60?180/yr.

194 3. ****Capture-quality variance = the failure mode**** (gym lighting, off-center angle,
195 occlusion, mixed resolutions). **Validate:** benchmark rally-detection P/R on 100+ real
196 amateur clips; define a "good-capture" spec + auto-reject (ties to
197 `VIDEO\_RECORDING\_GUIDELINES.md` + the observability reports).

198 4. ****Monetization mismatch in best-participation markets**** (India/Indonesia/China are
199 price-sensitive, hard to bill). **Validate:** local pricing/payments pilot; per-academy
200 INR licensing.

201 5. ****Competitive response**** (SwingVision or a team-sports incumbent adds racket sports).
202 **Validate:** ship a badminton-quality advantage incumbents can't trivially match
203 (offline self-improving detection, per-rally tagging).

204 6. ****B2B sales-cycle drag**** for clubs/federations/schools. **Validate:** single-coach/
205 academy self-serve before institutional deals.

206 7. ****Clean-licensing positioning**** matters mainly to B2B buyers ? confirm it's a real
207 purchasing criterion via 5?10 buyer interviews.

208

209 ****Cheapest next steps:**** (a) 10?15 discovery interviews with Indian badminton academy
210 owners/coaches; (b) a capture-quality benchmark on real amateur footage; (c) a fake-door
211 pricing test across B2C (~USD 99/yr) and B2B academy (~USD 600/yr).

212

213 ---

214

215 ## 7. Sources

216

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[https://sgbonline.com/itf-global-tennis-report-2021-shows-worldwide-participation-](https://sgbonline.com/itf-global-tennis-report-2021-shows-worldwide-participation-gains/)
234 gains/

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236 [in-every-age-group-and-region-for-third-straight-year-significant-investments-still-needed-for-](https://www.thedinkpickleball.com/report-pickleball-participation-surges-to-nearly-20-million-in-2024/)
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[https://www.thedinkpickleball.com/report-pickleball-participation-surges-to-](https://www.thedinkpickleball.com/report-pickleball-participation-surges-to-nearly-20-million-in-2024/)
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242 ;
<https://www.pixellot.tv/> ; <https://www.hudl.com/pricing> ;

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244

18. user (2026-06-21T03:17:42.956Z)

```
1      # Internationalization (i18n) ? Kannada-first, by design
2
3      **Status:** Design (P0 deliverable; phases P1+ are owner-gated, not started).
4      **Owner decision required** before any code lands ? see [Open decisions](#open-
decisions-for-the-owner).
5
6      ---
7
8      ## 1. Context ? why this, why now
9
10     The PMF beachhead is **Bangalore badminton academies**
([`PMF_MARKET_RESEARCH.md`](PMF_MARKET_RESEARCH.md)).
11     Many academy coaches, parents, and players are far more comfortable in **Kannada** than
English.
12     A product that only speaks English adds friction exactly at the point of first contact.
So the
13     product must speak **Kannada first**, **Hindi next**, and extend to other Indian
languages
14     (Tamil, Telugu, Marathi, ?) **without a refactor**.
15
16     The owner's directive is precise: make the product **internationalizable _by design**.
Adding a
17     language must be **"add a reviewed catalog" ? not "go re-touch the code."** The
translations
18     themselves can wait; **the foundation that makes them cheap to add is the actual
deliverable.**
19
20     Today there is **zero i18n** in the repo (no i18n/locale/gettext/babel anywhere). Every
21     user-facing string is hardcoded English across four surfaces:
22
23     | Surface | Where | Examples (today) |
24     |---|---|---|
25     | **UI chrome** | `backend/api/static/{index.html,app.js,style.css}` (~40?50 strings) |
`"Validate & Index Video"`, `alert("Validation Failed! ?")`, `` `${count} Rally Segment${count}
> 1 ? 's' : '' } Selected` `` |
26     | **Backend coach-facing text** | `backend/main.py` `HTTPException(detail=?)`;
`backend/pipeline/validators/` `` `message`; `backend/reporting/spec.yaml` `` `customer_verdict` +
per-rule `customer_message` | `Video file not found at '?'`; `Video quality rejected: Focus
check failed?`; ``? Your highlights are ready.` `` |
27     | **Self-navigating NL queries** (owner ask, future) | not built yet | **"show me all
rallies > 15 shots where I won" ? the **query and its answer** must work in the user's language
|
28     | **Coach-mode recommendations** (owner ask, future) | not built yet | localized
recommendation text, not English |
29
30     A React/Vite SPA is planned to replace the static UI ([`UI_ALPHA_EXECUTION_PLAN.md`](
ADR-009))
31     in a **separate private repo `avidullu/khelsutra`**. The design below is therefore
deliberately
32     **framework-agnostic**: the catalog format and locale-negotiation contract are shared,
so the
33     static UI today and the SPA tomorrow consume the *same* translation source.
34
35     **Scope confirmed with owner:** UI + coach-facing backend text + (designed-for, built-
later)
36     NL/coach features. **Translations are Gemini-drafted, human-reviewed** ? this is
*inference*
37     use of a paid LLM, which the [licensing guardrail](#guardrail-alignment) explicitly
allows
```

```

38     (it is *not* a training-data source). The owner reviews the first draft of every string.
39
40     ---
41
42     ## 2. Core architecture
43
44     ### 2.1 The "by design" invariant ? no hardcoded user-facing strings
45
46     Every coach-facing string is a **stable key** resolved at render time against the active
47     locale.
48     Code references `t("compile.button")`, never the English prose. This is the single rule
49     that
50     makes "add a language = add a catalog" true, and it is **enforced by CI** (§2.6) so it
51     cannot
52     silently regress as the codebase grows.
53
54     ```
55     "Validate & Index Video"                ? t("video.action.validate_index")
56     alert("Validation Failed! ?review the checklist") ? t("toast.validation_failed")
57     `${count} Rally Segment${count>1?'s':''} Selected` ? t("compile.selected", {count}) #
58     ICU plural
59     HTTPException(detail="Video file not found at ?")? detail = tr(lang,
60     "error.video_not_found", {path})
61     ```
62
63     ### 2.2 One canonical English source catalog, per-locale siblings
64
65     ```
66     locales/
67     en/  common.json  # source of truth ? devs author keys here
68     kn/  common.json  # Kannada  (Gemini-draft ? human-review)
69     hi/  common.json  # Hindi    (next; first exercise of the extensibility checklist)
70     ```
71
72     - **Format: i18next-flavored JSON** ? works for the vanilla shim today *and*
73     react-i18next in
74     the SPA, with no conversion.
75     - **ICU MessageFormat** for interpolation and plurals: `{count, plural, one {# rally}
76     other {# rallies}}`.
77     This matters immediately ? the current UI already hand-rolls English plurals
78     (`Rally Segment${count > 1 ? 's' : ''}`), and Kannada/Hindi plural rules differ from
79     English.
80     ICU encodes each language's rules in the catalog, not the code.
81     - `en` is the source of truth; **every other locale is keyed identically** (enforced by
82     the
83     key-parity check in §2.6).
84
85     ### 2.3 Locale-negotiation contract (shared frontend ? backend)
86
87     One resolution order, defined **in one place** and honored by both UI and backend:
88
89     ```
90     persisted user preference ? UI switcher / ?lang=<code> ? Accept-Language header ?
91     default "en"
92     ```
93
94     - Supported locales are enumerated in **a single module** (one list to grow). Anything
95     outside
96     it falls back to `en`.
97     - The backend localizes its own responses (`HTTPException.detail`, report `customer_`)
98     from the
99     same `lang` query param / `Accept-Language` header, so a Kannada UI gets Kannada
100     errors.
101     - **Always-fallback to `en`** for any missing key ? a missing translation must degrade
102     to

```

```

89     English, never to a blank or a raw key in front of a user.
90
91     ### 2.4 Backend text ? keys + params, not prose
92
93     Validator `message`s and API errors today bake in English prose *and* interpolated
values
94     (`f"?sharpness score of {avg} is below ? threshold of {threshold}."`). The design
separates
95     the two:
96
97     - The validator/endpoint emits a **stable code + a params dict**
(`("blur.below_threshold",
98     {"score": avg, "threshold": threshold})).
99     - The **prose lives in the catalog**, looked up at the API boundary in the negotiated
locale.
100    - `ValidationResult` keeps an English `message` for logs/back-compat, but gains an
optional
101    `message_key` + `message_params`; the report/serializer prefers the key when present.
This
102    keeps internal/technical messages English (good for debugging) while customer-facing
text
103    localizes.
104
105    `spec.yaml` `customer_verdict` and per-rule `customer_message` become **keyed** (or per-
locale
106    overlays keyed by rule `code`); the technical `message` blocks stay English (they cite
107    `README#Qn` and are for the operator, not the end user).
108
109    ### 2.5 Indic font strategy ? (open decision ? see §4)
110
111    The bundled UI/watermark fonts (Outfit, Roboto Slab) **do not cover Kannada or
Devanagari**.
112    **Noto Sans Kannada / Noto Sans Devanagari are OFL-licensed**, which is the same tension
we hit
113    with the watermark font: OFL is not in the strict **MIT/Apache/BSD** code/data/weights
guardrail.
114    Fonts are *media assets*, not code or model weights, and OFL is explicitly designed for
115    unrestricted redistribution ? so the cleanest path is to **carve out an explicit OFL-
for-fonts
116    exception**, the same way we'd treat any bundled media. Options put to the owner in §4.
117
118    Note the **video watermark already takes a `font_path`** (`WatermarkConfig.font_path`,
119    `backend/pipeline/stitcher/compiler.py`), so stamping a Kannada/Hindi watermark is a
120    *configuration* change (point it at a Noto font) once the font decision is made ? no new
code.
121
122    ### 2.6 CI enforcement (the guardrail that keeps it "by design")
123
124    Two checks, both pure-Python and runnable in this repo (no GPU/torch needed):
125
126    1. **Key-parity test** ? every locale catalog has *exactly* the `en` key set; fail on
missing
127    keys, extra keys, or untranslated placeholder markers (e.g. a `"_MISSING_"`
sentinel left
128    by the draft script). This is also the **extensibility proof**: drop a stub `ta.json`
with
129    one key and CI immediately lists every gap.
130    2. **No-bare-strings lint** ? a check that flags user-facing string literals not routed
through
131    `t()` / the backend `tr()` (scoped to the UI handlers and the API/validator/report
text
132    paths). Starts as a curated allowlist so it can land without boiling the ocean.
133
134    ### 2.7 Translation pipeline
135

```

```

136     ``
137     dev authors en keys ? scripts/i18n_draft.py (Gemini, INFERENCE only) ? kn/hi draft
138                               ? native-speaker review (owner reviews first draft)
139                               ? CI key-parity + placeholder check ? merge
140     ``
141
142     `scripts/i18n_draft.py` reads `locales/en/*.json`, asks Gemini for each missing key in
the
143     target locale **with the key name and any ICU placeholders preserved**, and writes a
draft
144     catalog with every machine-drafted value flagged for review. It never touches `en`,
never
145     invents keys, and is a *drafting aid* ? a human reviews before merge. (Reuses the
existing
146     Gemini provider credentials/config; this is the same "paid LLM = inference, never
training"
147     posture as the rest of the system.)
148
149     ---
150
151     ## 3. Phased execution (owner-gated; ONE PR per slice, off `master`)
152
153     | Phase | Scope | Repo | Gate |
154     |---|---|---|---|
155     | **P0** | **This design doc** (`docs/I18N_PLAN.md`) | this | _the immediate
deliverable_ |
156     | **P1** | Foundation + **static-UI retrofit**: `locales/` + `en` source catalog
(extract the ~40?50 UI strings ? keys), a tiny vanilla **t(key, vars)** shim + `data-i18n`
wiring in `index.html`/`app.js`, **language switcher (en/kn)** + `localStorage` persistence +
`?lang=`, the font decision applied, and the **kn** catalog (Gemini-draft ? owner review).
Reuse the existing static mount / `serve_index` in `backend/main.py`. | this | owner approval |
157     | **P2** | **Backend + report localization**: `backend/i18n/` with `tr(lang, key,
params)` (same catalog format), refactor validator `message`s to keys+params, localize
`HTTPException.detail` + `spec.yaml` `customer_verdict`/`customer_message`, wire `Accept-
Language`/`?lang=`. | this | owner approval |
158     | **P3** | **SPA adoption**: **react-i18next** consuming the *same* `locales/` catalogs;
documented FE?BE contract. | `avidullu/khelsutra` | needs repo access; or owner/Opus executes
there |
159     | **P4** | **Conversational + coach-mode** (future): locale-aware NL query intake ?
**templated localized answers**; coach-mode recommendation **templates**. Ties to the owned-
model / PLATFORM $5A + rich-metadata work. | this + model | gated on those features existing |
160
161     **Extensibility checklist** (this *is* the by-design contract ? runs the same every
time):
162     > add `locales/<code>/common.json` ? `scripts/i18n_draft.py --locale <code>` ? human
review ?
163     > add `<code>` to the supported-locales list ? confirm font coverage ? CI key-parity
green.
164
165     **Hindi (`hi`) is the first real exercise of this checklist** ? if adding `hi` requires
touching
166     anything other than the steps above, the design failed and we fix the design, not paper
over it.
167
168     ---
169
170     ## 4. Open decisions (for the owner)
171
172     1. **Target-UI scope** ? confirm the *shared framework-agnostic foundation* (prove on
the static
173     UI now, SPA adopts the same catalogs later). Alternative: scope only to the SPA and
skip the
174     static-UI retrofit. _Recommendation: shared foundation ? it's the cheap, low-risk way
to
175     validate the catalog/negotiation contract before the SPA exists._

```



```

176      2. **Indic font licensing** ? (a) **allow OFL for bundled fonts** and self-host Noto
Sans
177      Kannada/Devanagari _(recommended; treats fonts as media, not code/weights); (b) rely
on
178      **system Indic fonts** (no bundle, inconsistent rendering across machines); (c)
**Google Fonts
179      CSS** (runtime external dependency, privacy/offline concerns for a local-first
product). This
180      also unblocks the **Kannada/Hindi video watermark** (needs a Noto `font_path`).
181      3. **v1 locale set & fallback** ? confirm `en` + `kn` for v1, `hi` next; always-fallback
to `en`
182      for missing keys.
183      4. **Backend-localization depth for v1** ? recommendation: localize the **customer
verdict /
184      report `customer_`** and **validation rejection reasons** first (highest user
impact);
185      defer technical/operator messages (they stay English by design).
186
187      ---
188
189      ## 5. Key files / seams
190
191      - **Localize:** `backend/api/static/{index.html,app.js,style.css}`; `backend/main.py`
192      (`HTTPException.detail` + the `serve_index`/static mount);
`backend/pipeline/validators/`
193      (`message` ? key+params; see `validators/base.py::ValidationResult`);
194      `backend/reporting/spec.yaml` (`customer_verdict` + per-rule `customer_message`).
195      - **New:** `locales/{en,kn,hi}/`, `backend/i18n/` (`tr()` + supported-locales +
negotiation),
196      `scripts/i18n_draft.py`, a CI key-parity test under `tests/` + a no-bare-strings check
wired
197      into `.github/workflows/ci.yml`.
198      - **Fonts:** reuse the bundled-asset pattern in `backend/assets/fonts/` (Apache-2.0
Roboto Slab
199      ships there today); add Noto Indic fonts there **iff** decision §4.2(a).
200      - **SPA:** `avidullu/khelsutra` (react-i18next) ? separate repo.
201
202      ---
203
204      ## 6. Verification
205
206      - **CI key-parity test** (Python, this repo): every locale has every `en` key; fail on
207      missing/extra/placeholder. Drop a stub `ta.json` ? CI flags every gap (the
extensibility proof).
208      - **Static UI:** load `?lang=kn` ? rendered strings resolve from the `kn` catalog;
switcher
209      persists across reload; any missing key falls back to English (no raw keys shown).
210      - **Backend:** request with `Accept-Language: kn` (or `?lang=kn`) ? localized error
`detail` +
211      report `customer_`; unit tests for fallback, ICU interpolation, and plural rules.
212      - **Fonts:** Kannada + Devanagari render correctly ? **manual eyeball on the owner's
machine**
213      (the cloud container can't render fonts).
214      - **Docs-only P0 PR:** internal links resolve; indexed in [`docs/README.md`](README.md);
215      cross-linked from [`PLATFORM_ARCHITECTURE.md`](PLATFORM_ARCHITECTURE.md) (the NL/coach
surfaces)
216      and [`COMMERCIALIZATION.md`](COMMERCIALIZATION.md) (the Bangalore beachhead).
217
218      ---
219
220      ## Guardrail alignment
221
222      - **Paid LLM = inference only.** Gemini drafts translations (inference); it is **never**
a
223      training-data source. ? consistent with [`CLAUDE.md`](../CLAUDE.md) and the ratified

```

```

guardrail.
224 - **Licensing.** Code stays MIT/Apache-2.0/BSD. The *only* tension is **fonts** (OFL) ?
handled
225 as an explicit, owner-approved media-asset exception in §4.2, not a silent dependency.
226 - **No minors / biometrics scope creep.** i18n touches presentation only; it introduces
no new
227 personal-data collection.
228
229 [ `UI_ALPHA_EXECUTION_PLAN.md` ]: ./UI_ALPHA_EXECUTION_PLAN.md
230

```

19. assistant (2026-06-21T03:17:46.497Z)

I now have all the source material. Let me also check the next-session UX/localization handoff memory note referenced in MEMORY.md, and the VIDEO_RECORDING_GUIDELINES (referenced for recording-education-first), to ensure the brief is faithful and complete.

20. user (2026-06-21T03:17:47.461Z)

```

1 ---
2 name: next-session-ux-localization
3 description: Consolidated handoff/plan for the NEXT session ? focus SOLELY on (A) making
the UI super user-friendly for non-tech-savvy users and (B) localizing in Hindi + Kannada.
Grounded in existing repo docs (owner directive 2026-06-21).
4 metadata:
5   node_type: memory
6   type: project
7   originSessionId: 577fd4df-ed3f-43d3-854d-d63d732e2e07
8 ---
9
10 **Owner directive (2026-06-21, session wrap):** the next session focuses **SOLELY on
making the product super user-friendly (non-tech-savvy) AND localized (Hindi + Kannada)**.
*Research first using the existing repo docs, THEN craft the plan* (don't reinvent ? there's
prior design). The end goal: **a non-tech-savvy coach/parent reads the page, confidently sets up
recording, runs the tool, and gets highlights ? in their own language.**
11
12 ## Clean slate at handoff (2026-06-21)
13 Both repos on `master`, **0 open PRs**, no orphaned worktrees, **no GCP VMs** (no
billing). khelsutra master `5de981f`; engine (`Khelsutra/badminton-highlight-indexer`) master
`4134721` (DECOUPLED_COMPUTE archived as delivered). Engine local checkout is the **concurrent
session's** branch ? don't disturb it; use an isolated `git worktree` off `origin/master` for
any engine work (the #116/#117 collision rule).
14
15 ## ? Ramp-up kit (READ THESE before crafting the plan ? the owner's "use the docs" ask)
16 **Localization (the pillar):** **`engine docs/I18N_PLAN.md`** ? Kannada-first?Hindi-
next, **by design** (no hardcoded strings; `t("key")` + ICU MessageFormat plurals; one `en`
source catalog + per-locale siblings; locale-negotiation order = persisted pref ?
switcher/?`lang=` ? `Accept-Language` ? `en` fallback; **Gemini-DRAFT?human-REVIEW** [inference,
never training ? guardrail-OK]; CI **key-parity** + **no-bare-strings** lint; **Noto Sans
Kannada/Devanagari**, OFL font decision). Phases P0?(doc)?P1(static-UI retrofit)?P2(backend
text)?P3(**SPA react-i18next**)?P4(conversational/coach-mode).
17 **UX + personas (the pillar):** **`engine docs/UI_PROPOSAL.md`** §2 personas
(Coach/academy = primary B2B India, Kannada-comfortable, records on static cameras; Enthusiast
B2C) + the "teach recording BEFORE upload / zero hand-holding" principle + the 6 alpha screens
(A1 Matches · A2 Upload+recording-checklist · A3 Job progress · A4 Rally review ? · A5
Highlights · A6 Report).
18 **Non-tech-savvy reality:** **`engine docs/PMF_FIELD_SIGNALS.md`** +
**`docs/PRODUCT_FIELD_ASSOCIATE_FLYER.md`** (field associate works in **Kannada/Hindi/English**;
coaches/players/**parents**; "simple summary"; **WhatsApp-native**; possible B2C-to-parents
pivot). **`docs/PERSONALIZATION_PLAN.md`** (coach overlay).
**`docs/KHELSUTRA_PRODUCT_OVERVIEW.md`** (who it's for).
19 **Current product surfaces (what to localize + simplify):** khelsutra **`web/` SPA**
(per-rally operator console ? `docs/PER_RALLY_SELECTION.md`) + **`site/`** marketing (homepage,
`/capture` checklist, `own-model`, request-access form, post-signup reveal). `web/README.md`.
20

```

```

21     ## ? THE adaptation the next session must make
22     `I18N_PLAN.md` was written for the OLD static UI (`backend/api/static/`). The
product is now the khelsutra web/ SPA (Vite/React/TS/Tailwind) + the static marketing
site/ (vanilla HTML). So adapt the design's framework-agnostic foundation to:
23     - web/ SPA ? react-i18next (the doc's P3) consuming `locales/{en,kn,hi}/.json`
catalogs (ICU plurals); a language switcher + persistence; localize the per-rally UI (query
examples/pills, rally cards, footer, build/compile/download, errors). Honest scope still applies
(duration-first; shot-count experimental).
24     - Marketing site/ (vanilla) ? a tiny `t(key,vars)` shim + `data-i18n` over
`index.html`/`capture.html`/`own-model.html` (no framework), sharing the SAME catalog format.
The form + reveal copy + the recording gist localize too.
25     - Backend (engine) ? P2 (errors/report `customer_` as keys+params) is owner-gated +
a separate worktree PR.
26
27     ## Two threads to plan (one tracked project doc each, per `[[working-agreement-tracked-
project-docs]]`)
28     (A) Non-tech-savvy UX ? research ? plan. Likely levers (validate against the docs +
personas): plain-language/low-jargon copy, fewer steps + bigger touch targets, mobile-first
(coaches use phones), visual/iconographic over text, recording-education-first (? already
shipped the homepage "Record it right" gist + `capture`), WhatsApp-friendly share, a guided
"upload ? wait ? here's your reel" happy path with zero knobs. The per-rally query bar already
leads with duration and flags shot-count "experimental" ? keep simplifying.
29     (B) Hindi + Kannada localization ? adapt `I18N_PLAN.md` to `web/` + `site/`; stand
up the catalog + negotiation + a first `kn` (then `hi`) draft via `scripts/i18n_draft.py`-style
Gemini-draft?owner-review; CI key-parity; Noto fonts.
30
31     ## Open OWNER decisions to resolve first (from I18N_PLAN §4)
32     1. Indic font licensing ? recommend (a) OFL exception for bundled fonts, self-
host Noto Sans Kannada/Devanagari (treats fonts as media, not code/weights ? also unblocks a
Kannada/Hindi reel watermark via the existing `WatermarkConfig.font_path`).
33     2. v1 locale set ? recommend `en` + kn for v1, hi next; always-fallback
to `en`.
34     3. Backend-localization depth ? recommend customer verdict / report `customer_` +
validation-rejection reasons first; operator/technical text stays English.
35     4. UX scope ? which surface first: the marketing `site/` (first-contact friction)
and/or the `web/ SPA` (the operator console).
36
37     ## Already shipped this session (DON'T redo ? build on it)
38     Per-rally MVP P1?P5 (query bar/grid/footer/build?compile?download, honest duration-
first, shot-count parked experimental); engine M1 (`/api/health` + compile `download_url`);
marketing site ? post-signup reveal (tool screencast + 30s reel, watchable: manual close +
holds-while-playing), submit "Sending?" feedback, "Record it right" homepage gist (from
`VIDEO_RECORDING_GUIDELINES.md`) + `capture`; behavioral jsdom tests (test-only dep;
runtime stays dependency-free); worker observability fix (logs a 0-segment cold-start,
[#271]). Promo/CUJ screencasts in `gs://khelsutra-rally-corpus/promo/`. ? Cold-start blocker
still open: native-WASB `worker infer` config that reproduces the engine session's 22-rally
result (see `[[current-state]]`). `[[monetization-plan]]` `[[khelsutra-domain-launch]]`
39

```

21. user (2026-06-21T03:17:47.784Z)

```

1     # Video Recording Guidelines (DRAFT ? thresholds pending empirical data)
2
3     Status ? Draft / placeholder . Date 2026-05-30
4     Why this exists: highlight quality is bounded by input quality. To get reliably
good auto-generated highlights, the source video must be recorded a certain way (camera height,
court position, resolution, frame rate, stability, framing). We want to (1) derive the
minimum acceptable capture spec empirically from the precision/recall of the solutions we
build, and (2) publish user-facing best practices so a user knows how to record footage that
the product can index well.
5
6     > This is a tracked thought, not a finished standard. The exact numeric thresholds
below are placeholders to be replaced with values measured against our golden set. See "How
we'll set the real numbers" (§3). Captured now (per the team's working agreement) so it isn't
lost; to be revisited.

```

```

7      >
8      > **Update (2026-06-02):** the **frame-rate** and **camera-geometry** rows now carry
**literature-backed starting values** (cited inline). They remain *starting points* to confirm
empirically per §3. The golden-set **corpus** diversity spec (what footage to collect, not just
how users should record) is now in [GOLDEN_SET_PHASE1_VIDEOS.md](GOLDEN_SET_PHASE1_VIDEOS.md).
9      >
10     > **Update (2026-06-07):** **AUDIO is now a first-class capture requirement** (new row
below), per ADR-007 ? we will use the shuttle "thwack" as a rally cue (audio+visual fusion lifts
recall). This is a hard "record with audio on, mic unobstructed" ask, not a placeholder;
efficacy on real GoPro audio is being validated (experiment E1 in
[AUDIO_RALLY_DETECTION_PLAN.md](archives/research/AUDIO_RALLY_DETECTION_PLAN.md)).
11
12     ---
13
14     ## ?? THE #1 CAPTURE RULE: ONE MATCH PER FRAME (single court)
15
16     > **For the best auto-highlights, record a SINGLE match on a SINGLE court ? with no
other matches
17     > playing in the background or on adjacent courts.** This is the happy path the product
is tuned for,
18     > and as of **2026-06-18 it is MEASURED, not a guess:**
19     >
20     > - On a golden **multi-court** clip, the rally detector fired **87 "rallies" against 43
real ones ?
21     > precision ? 0.20 (~80% false positives)**, almost all of them background-court games
it mistook
22     > for play.
23     > - On **single-court** clips, precision is markedly higher and rally boundaries are
tight; local even
24     > **out-recalls Gemini** on normal single-match play.
25     >
26     > **Multi-court is a KNOWN BOTTLENECK we are actively working to fix** (player-state +
court-occupancy
27     > signals ? see [RALLY_DETECTION_GAP_CLOSURE.md](RALLY_DETECTION_GAP_CLOSURE.md)). Until
that lands,
28     > **single-match capture is strongly recommended** ? a reel made from busy multi-court
footage *will*
29     > include false "rallies" from the neighbouring games. **North star (for now):
consistently beat Gemini
30     > on the single-match happy path.**
31
32     ---
33
34     ## 1. Why input capture matters (grounded in what we already know)
35
36     ADR-004 found a concrete, capture-dependent failure mode: on **broadcast** footage the
camera *tracks the players*, so player pixel-motion dips during rallies and spikes on
pans/replays ? making player-motion a poor rally proxy. The lesson generalizes: **the pipeline's
accuracy is conditional on how the video was shot.** A **static, full-court** camera (typical of
a fixed tripod recording) behaves very differently from broadcast ? and is likely much
friendlier to both the motion/detector stages and the shuttle-tracking substrate (WASB/TrackNet,
ADR-001).
37
38     So capture guidance isn't cosmetic ? it directly moves precision/recall.
39
40     ---
41
42     ## 2. The dimensions that likely matter (hypotheses to test)
43
44     | Dimension | Why it plausibly affects P/R | Placeholder guidance (UNVERIFIED) |
45     |---|---|---|
46     | **Resolution** | Shuttle is tiny & fast; too few pixels ? detector misses it | ?
**1080p** preferred; **720p** likely floor; 2K/4K may help shuttle tracking but ? compute/cost |
47     | **Frame rate** | Motion blur; boundary precision; fast-ball detection | ? **30 fps**
badminton/pickleball; **? 60 fps table tennis** ? 30 fps collapses fast-ball detection (**22.5%

```

? 93.2%** at 60 fps; 120 ? 60, [Nature Sci Rep](https://www.nature.com/articles/s41598-023-42966-6)) |

48 | **Camera position** | Static full-court view avoids the broadcast camera-tracking trap (ADR-004) | **Fixed tripod**, elevated, behind/above one baseline, whole court in frame |

49 | **Camera height & angle** | Occlusion of players/shuttle; court keystoneing | **3.75 m** elevated (~4.5 m), **~30° down-tilt**, behind one baseline, whole court + net headroom ([badminton-CV setup](https://pmc.ncbi.nlm.nih.gov/articles/PMC10747833/)) |

50 | **Framing / margins** | Cropping the court or players breaks detection & relevance check | Entire court + a margin; players never leave frame |

51 | **Stability** | Handheld jitter inflates motion signals | Tripod / fixed mount; no panning or zoom during play |

52 | **Lighting / exposure** | Blur & contrast affect detection and the existing blur/relevance validators | Even indoor lighting; avoid backlight/glare; minimize motion blur |

53 | **Single match in frame (no background games) ? ? MEASURED, the #1 rule** | Adjacent/background-court games get detected as false rallies ? this is the **dominant precision failure** | **One match per frame, single court**. Multi-court precision measured at **0.20** (vs 0.31 single, tighter boundaries); multi-court fix in progress ? see RALLY_DETECTION_GAP_CLOSURE.md |

54 | **Court contrast / background** | Relevance check uses court color; busy crowds add false motion | Clear court markings; minimal moving background where possible |

55 | **Codec / container** | Pipeline already validates MP4 + decodes H.264/HEVC | MP4 (H.264/HEVC) as today; align with `archives/MONETIZATION\_AUDIT.md` codec notes |

56 | **Audio (REQUIRED ? ADR-007)** | Shuttle "thwack" hits are a strong rally cue; audio+visual fusion lifts rally *recall* (the WASB bottleneck) ? see ADR-007 / archives/research/AUDIO_RALLY_DETECTION_PLAN.md | **Record with AUDIO ENABLED** (GoPro default ? do not mute). Keep the **mic unobstructed**; **avoid aggressive wind/noise reduction** that flattens the sharp impact transient. Camera reasonably close to the court helps SNR (shuttle hits are quiet). |

57

58 These map onto checks that **already exist** in `backend/pipeline/validators/` (resolution, fps, blur, scene-cut, court relevance) ? so several can be enforced/measured programmatically rather than only advised.

59

60 ---

61

62 **## 3. How we'll set the REAL numbers (empirical plan)**

63

64 Replace every placeholder above with a value derived from data, using the golden-set harness:

65

66 1. Take golden-set clips (GOLDEN_SET_VENDORING.md) spanning a **range** of each dimension ? e.g. the same match re-encoded at 480p/720p/1080p/4K, or clips at different camera heights.

67 2. Run the production pipeline; score with `backend/eval/metrics.py`.

68 3. Plot **precision/recall vs. each dimension** ? find the knee where quality collapses.

69 4. Set the **published minimum** at (knee + safety margin); set a **"recommended"** tier above it.

70 5. Re-derive when the pipeline changes materially (new detector/segmenter shifts the knees).

71

72 This turns "record in 1080p" from an assertion into a measured threshold ? and lets us tell users ***why***.

73

74 ---

75

76 **## 4. Two surfaces for the output**

77

78 1. **User-facing best-practices guide** ? short, friendly, illustrated: "Put the camera here, this high, record at ?X, keep it still." The goal is that a user can record once and get great results.

79 2. **Programmatic pre-flight check** ? extend `backend/pipeline/validators/` to **reject or warn** on out-of-spec uploads at ingest (resolution/fps/blur/framing), so the product fails fast with actionable feedback instead of producing poor highlights silently.

80

81 ---

82
83 ## 5. Open questions
84 1. Which dimension dominates ? resolution, camera staticness, or height? (Run the
ablations in §3 to rank them.)
85 2. Do we **enforce** minimums (reject uploads) or only **warn**? Likely warn first,
enforce once thresholds are trustworthy.
86 3. Single fixed-camera spec, or several supported "recording profiles" (e.g. baseline-
elevated vs. side-elevated)?
87 4. How does the owned/licensed golden corpus need to grow to **cover** the capture
conditions real users will actually use? (Feeds GOLDEN_SET_VENDORING.md §8.)
88

22. assistant (2026-06-21T03:19:39.702Z)

I have read all the specified docs completely plus the directly relevant companions (I18N_PLAN, VIDEO_RECORDING_GUIDELINES, the next-session handoff). I have enough to produce the exhaustive, faithful brief.

One note for the engineer: the `git pull --ff-only` failed because the current branch `docs/doc-status-register` is local-only (no matching remote ref), so I worked from the local checkout; nothing was force-merged. All findings below are grounded strictly in the docs cited ? no invention.

Ramp-up Brief: KhelSutra Users + Non-Tech-Savvy UX

Audience: an engineer making KhelSutra friendly to non-tech-savvy coaches/parents in Bangalore.

Sources (all faithful, nothing invented): ``docs/UI_PROPOSAL.md``, ``docs/PMF_FIELD_SIGNALS.md``, ``docs/PMF_MARKET_RESEARCH.md``, ``docs/KHELSUTRA_PRODUCT_OVERVIEW.md``, ``docs/PRODUCT_FIELD_ASSOCIATE_FLYER.md``, ``docs/FIELD_VISIT_PLAYBOOK.md``, ``docs/FIELD_VISIT_ANSWER_SHEET.md``, ``docs/PERSONALIZATION_PLAN.md``, ``docs/UI_ALPHA_EXECUTION_PLAN.md``, plus directly-cited companions ``docs/I18N_PLAN.md`` and ``docs/VIDEO_RECORDING_GUIDELINES.md``.

Repo-freshness caveat: the local branch ``docs/doc-status-register`` has no remote, so ``git pull --ff-only`` could not run; this brief reflects the local checkout (may lag remote master). Nothing was merged/forced.

1. What KhelSutra is (one breath, for context)

KhelSutra turns ordinary badminton video (phone/GoPro on a tripod behind the court) into useful pieces **automatically**: it finds the rallies, stitches a **highlight reel** (MP4 with a small **"Generated by Khelsutra.guru"** mark), and gives a **simple summary** (how many rallies, how much real play time). Core loop: **record ? upload ? get back a reel + summary**. No editing skill, no special hardware, no manual scrubbing. (KHELSUTRA_PRODUCT_OVERVIEW "In one line"; "What it does today.")

The hard, differentiating problem is **multi-court** halls (busy/noisy footage); it already works very well on **clean single-court** recordings and multi-court is the sharpening focus. (OVERVIEW "the problem we're solving"; VIDEO_RECORDING_GUIDELINES §"#1 capture rule": multi-court precision measured ?0.20 vs ?0.31 single.)

2. Personas

PRIMARY ? Coach / academy (B2B, India / Bengaluru)

- **Who:** academy owner / head coach ? **the person who decides what the academy spends money on** (FIELD_VISIT_PLAYBOOK "Who to talk to"). Has batches of match recordings; wants rally review, per-player reels for students/parents, per-rally stats (UI_PROPOSAL §2).

- **Already records** on static/fixed cameras (tripod/CCTV/phone rig) ? this is the behavior the product depends on and the reason academies are the cleanest PMF fit (UI_PROPOSAL §2; PMF_MARKET_RESEARCH §3 "buyer, recorder, payer aligned"; PMF_FIELD_SIGNALS H1).
- **Tech-comfort**: explicitly **non-tech-savvy** is the target ? the goal is "a non-tech-savvy coach/parent reads the page, confidently sets up recording, runs the tool, and gets highlights ? in their own language" (next-session handoff). The field associate needs **no tech background** and coaches are not assumed to either (FLYER "No tech background needed"; "No technical skills needed").
- **Device**: **phone-first**. Coaches use phones; the field associate carries the demo reel **saved offline on their phone** and shows it on a phone (PMF_FIELD_SIGNALS §4 demo asset; PLAYBOOK Phase-2). Note the alpha UI's stated compromise: **usable on phone, optimized for laptop** (UI_PROPOSAL §3 non-goals) ? i.e. mobile-first is a **gap to close**, not yet done.
- **Language comfort**: "Many academy coaches, parents, and players are far more comfortable in **Kannada** than English" (I18N_PLAN §1). The field associate works in **Kannada / Hindi / English**, choosing the mix per conversation (FLYER; PLAYBOOK; FIELD_ASSOCIATE flyer "Kannada / Hindi / English, as the conversation needs"). Today the app is **English-only** ("today the app is in English" ? OVERVIEW "How to use it"); Kannada-first/Hindi-next is the localization mandate (I18N_PLAN).

SECONDARY ? Enthusiast / serious player (B2C funnel)

- **Who**: records own matches; wants a highlight reel + basic stats; **shares with friends** (UI_PROPOSAL §2; OVERVIEW "serious players ? keep, and share"). Treated as a **funnel** to B2B, not the primary economic bet (PMF_MARKET_RESEARCH §3 "Volume play, weak economics; use as funnel").
- **Key adoption risk**: **tripod discipline** ? "the UI must teach recording guidelines **before** upload, not fail videos after" (UI_PROPOSAL §2).

EMERGENT THIRD ? Parents (B2C-to-parents)

- Parents are repeatedly named as an audience/consumer/payer: "per-player reels for students/**parents**" (UI_PROPOSAL §2); H3 audience includes "parents ask for clips" (PMF_FIELD_SIGNALS §2). A pre-registered **PIVOT C** exists: "payer is the parent, not the academy" ? re-test WTP as **B2C per-student reels** (?/reel to parents) with the academy as channel (PMF_FIELD_SIGNALS §3, triggered when H5 RED and Q3 names parents/players). The next-session directive explicitly scopes UX to "non-tech-savvy **coaches/parents**" (handoff).
- Parents are at least as non-tech-savvy and as phone-/WhatsApp-native as coaches ? design implication, not yet a built persona.

Personas the docs explicitly EXCLUDE / de-prioritize

- **Broadcasters/streamers** ? "Avoid: violates core constraint" (panning = failure mode) (PMF_MARKET_RESEARCH §3).
- **Minors** ? excluded from the training pool and never recorded for the library; coach-permitted **adult (18+)** students only in field work (FLYER ground rules; PLAYBOOK; PERSONALIZATION_PLAN; CLAUDE.md guardrail).
- The CLI/power-user/ops audience stays on the CLI ? "The CLI is not replaced" (UI_PROPOSAL §1).

3. Non-tech-savvy UX principles the docs assert

These are stated or directly implied by the docs ? each is real, not invented:

1. **Zero hand-holding / guided happy path with zero knobs**. Alpha goal is "10?25 invited testers, **end-to-end, zero hand-holding**" (UI_PROPOSAL §3). The happy path the handoff names: "upload ? wait ? here's your reel" with **zero knobs**; "a non-tech-savvy coach/parent reads the page, confidently sets up recording, runs the tool, and gets highlights" (next-session handoff). The OVERVIEW core loop is deliberately three steps: record ? upload ? reel + summary.
2. **Hide all technical machinery**. "Segmenter choice hidden"; the WASB/owned-model path is **not exposed** in alpha; alpha serves the Gemini path with no GPU in the request path (UI_PROPOSAL §1). The Report has a **"nerd view" toggle** so technical detail is opt-in only, default-hidden (UI_PROPOSAL §3, A6).
3. **Recording-education FIRST** (teach before upload, don't fail after). "The UI must teach recording guidelines **before** upload, not fail videos after" (UI_PROPOSAL §2). A2 carries an **inline recording-guidelines checklist** (pre-upload education from VIDEO_RECORDING_GUIDELINES) (UI_PROPOSAL A2). Already partly shipped: a homepage **"Record it right"** gist + a `/capture``

checklist page (next-session handoff "Already shipped"). The capture rules to surface plainly: fixed tripod, elevated, **whole court in frame**, landscape, keep it still (no pan/zoom), good lighting, **single match per frame**, **audio on/mic unobstructed** (VIDEO_RECORDING_GUIDELINES §2, §"#1 rule"; OVERVIEW "capture matters").

4. **Plain language / low jargon.** Product copy is deliberately plain: **"in plain words"** (FLYER); **"the simple summary"**, **"no editing skill needed"**, **"(plainly)"** framings throughout OVERVIEW. The handoff lever: **"plain-language/low-jargon copy"**.

5. **"Simple summary" as a first-class output.** Repeatedly named ? not "analytics dashboard" but **"a simple summary"** ? how many rallies were found and how much actual play time" (OVERVIEW; FLYER; PLAYBOOK; PMF_FIELD_SIGNALS). Keep the numbers minimal and human.

6. **Fewer steps + bigger touch targets + mobile-first.** Named levers in the handoff: **"fewer steps + bigger touch targets"**, **mobile-first** (coaches use phones). Today's alpha is only **"usable on phone, optimized for laptop"** (UI_PROPOSAL §3) ? closing the mobile gap is the work.

7. **Visual / iconographic over text.** Handoff lever: **"visual/iconographic over text"**. Status is already expressed as **chips/icons** (verdict ?/?/? per video) (UI_PROPOSAL A1). The capture guide is meant to be **"short, friendly, illustrated"** (VIDEO_RECORDING_GUIDELINES §4).

8. **WhatsApp-native sharing / feedback.** The field channel is WhatsApp (FLYER "phone/WhatsApp"; PLAYBOOK; UI_ALPHA_EXECUTION_PLAN feedback channel = "WhatsApp group or Google Form"). Handoff lever: **"WhatsApp-friendly share"**. Enthusiasts/parents want to **share** reels (UI_PROPOSAL §2; OVERVIEW). (Note: explicit share links are an alpha **non-goal** ? Beta Drop 3 adds athlete-facing expiring share links ? so "WhatsApp-friendly" today means the downloaded MP4 is easy to forward, per OVERVIEW "watch, download, and share.")

9. **Background processing the user can walk away from.** "It processes in the background. You can **close the tab and come back later**"; you don't have to sit and watch it" (OVERVIEW; backed by the async job model, B1). Sets the expectation that long Indian-uplink uploads + multi-minute processing are normal and don't require babysitting.

10. **Honest, calm expectation-setting (no overpromise).** See §5 (Copy/tone). This is a UX principle, not just copy: the product tells users plainly what it does **today** vs the roadmap, and frames misses as useful feedback, not breakage (OVERVIEW "Read this honestly" callouts).

11. **Correction is easy and low-stakes (the user is a co-builder).** A4 review must let a non-technical coach accept/reject/nudge with **bulk** "all good" and simple chips; "You don't need to be technical ? just tell us, in your words" (UI_PROPOSAL A4; OVERVIEW "What feedback helps us most").

12. **Localized first contact (Kannada-first, Hindi-next).** Reduce friction "exactly at the point of first contact" by speaking Kannada first; always **fall back to English** for any missing string so a user never sees a blank or raw key (I18N_PLAN §1, §2.3).

4. The 6 alpha screens ? what each must do, simply

(From UI_PROPOSAL §3 table; "simply" framing per the non-tech-savvy mandate.)

#	Screen	What it must do ? simply
A1	Matches (home)	Show the user's videos as a list, each with a single clear status chip (verdict ? / ?? / ?) and one obvious upload call-to-action. One glance = "what do I have, is it ready."
A2	Upload & Process	Teach recording first : an inline recording-guidelines checklist before/at upload (pre-upload education, not post-failure). Then drag-drop (chunked upload so big files on slow Indian uplinks work) ? auto-starts processing. No format/segmenter choices to make.
A3	Job progress	A plain stage list (validation ? segmentation ? AI confirm ? report) with live status, ending in a verdict banner . Reassures the user it's working and they can leave; sets the "background, come back later" expectation.
A4	Rally review	(the corpus engine) Video player + rally list; click a rally ? seek & play that span . Per-rally Accept / Reject / nudge start?end (±0.5s) ; a bulk "all good"; simple chips (duration, shot count + confidence). Must feel effortless for a non-technical coach ? every accept/reject/nudge silently produces the labels the model is starved for (and is the v1 "correction loop").
A5	Highlights	Select accepted rallies ? name the reel ? compile ? download ; list past reels. Output is a single shareable/downloadable MP4.
A6	Report	Render the customer-audience report (verdict, notes, FAQ refs) in plain language; a "nerd view" toggle hides the technical detail by default.

Backed by backend enablers B1?B7 (async job model #16, chunked upload, download endpoint, per-user scoping, retire buggy static UI, correction-loop `PATCH` endpoints, exposure hardening) ? see UI_PROPOSAL §3 dependency list. Sequencing/status in UI_ALPHA_EXECUTION_PLAN (PR-1 #87 and PR-2 #99 merged; PR-3 identity/exposure and PR-4 correction-loop pending; SPA M1?M4 pending). Note: A4's accept/reject/nudge depends on **PR-4 (B6)** landing.

Alpha non-goals (so the engineer doesn't over-build): no payments, no orgs/teams, no athlete profiles, no share links, no owned-model serving, no multi-sport selector (badminton only), no notifications, **no mobile-perfect layouts** (usable on phone, optimized for laptop), no conversational QA (UI_PROPOSAL §3).

5. Copy / tone guidance

- **Plain words, low jargon.** "The product, **in plain words**"; describe handling **(plainly)**; **simple summary**, **no editing skill needed** (FLYER; OVERVIEW). The handoff's first lever is "plain-language/low-jargon copy."
- **Honest, never overpromising.** Bright-line rule for field + product: **Honesty over positivity. Never oversell, never promise features, prices, or dates** (FLYER ground rule 1; PLAYBOOK ground rule 3). The OVERVIEW uses explicit **"Read this honestly"** callouts to separate **today** (rally detection + reels + play-time summary) from the **roadmap/direction** (shot detection, AI assistant coach, more sports, your language). Replicate that today-vs-direction split in product copy.
- **Frame misses as feedback, not breakage.** "Like any automatic tool, it can occasionally miss a real rally or include one that isn't ? telling us when that happens is **genuinely useful feedback**, not a sign it's broken" (OVERVIEW). A4/feedback copy should invite correction warmly.
- **"Show, never promise."** The demo reel is shown, results aren't talked up; the strongest artifact is **"this is what it looks like for you"** (PLAYBOOK ground rule 3; PMF_FIELD_SIGNALS §4; FLYER stretch goal). Lean on concrete sample output over claims.
- **Privacy stated plainly and reassuringly.** **"Your data stays yours ? and sharing is your choice"** ? opt-in, never automatic; **"Footage involving minors is never used ? and is never published"**; **"we never publish your footage"** (OVERVIEW; FAQ "Is my video private?"). Say it in plain language at the relevant moments (upload, share).
- **Don't quote accuracy numbers ? show a reel.** "The best answer is to **show a sample reel** rather than quote a number" (OVERVIEW FAQ "How accurate is it?").
- **Don't claim self-improvement that isn't on yet.** "today the tool ? does **not** retrain itself as you use it ? The 'gets sharper over time' ? is the model we're building toward ? **not a switch that's already on**" (OVERVIEW). Copy must not imply live personalization (consistent with PERSONALIZATION_PLAN: per-user *learning* is gated/largely unbuilt; only label-free baseline "batteries" + opt-in flywheel).
- **Pricing language: open, fair, undecided.** **"Pricing isn't set"** ? we're still learning what's fair." Never quote a price; ask "what would feel fair to you?" (OVERVIEW; PLAYBOOK ground rule 1).
- **Localized tone, not just translated strings.** Kannada-first reduces first-contact friction; translations are **Gemini-drafted, human-reviewed** (owner reviews first draft of every string), with ICU plurals so Kannada/Hindi plural rules are correct, and always-fallback to English (I18N_PLAN §1, §2.2?2.7). Watermark can be localized via the existing `WatermarkConfig.font\_path` once the Noto-font decision lands.

6. UX levers, tagged by persona served

Each lever is asserted/implicit by the docs; tag = which persona(s) it primarily serves. (Coach = primary B2B coach/academy; Enthusiast = B2C player; Parent = B2C-to-parents; All = every non-tech-savvy user.)

#	UX lever (source)	Persona(s) served
L1	Recording-education BEFORE upload ? A2 inline checklist; homepage "Record it right" + `capture` (UI_PROPOSAL §2/A2; VIDEO_RECORDING_GUIDELINES §4; handoff)	Enthusiast (tripod-discipline is the adoption risk), All

| L2 | ****Guided happy path, zero knobs**** ? upload ? wait ? reel; auto-start processing; no segmenter/format choices (UI_PROPOSAL §1/A2; handoff) | ****All**** (esp. non-tech coach/parent) |

| L3 | ****Hide technical machinery**** ? segmenter choice hidden; "nerd view" off by default (UI_PROPOSAL §1, A6) | ****Coach, Parent**, All** |

| L4 | ****Plain-language / low-jargon copy**** (FLYER; OVERVIEW; handoff) | ****All**** |

| L5 | ****"Simple summary" as the headline output**** (OVERVIEW; FLYER; PMF) | ****Coach**** (teaching), ****Parent/Enthusiast**** (what happened) |

| L6 | ****Mobile-first, bigger touch targets, fewer steps**** (handoff; closes UI_PROPOSAL §3 "optimized for laptop" gap) | ****Coach**** (phone), ****Parent**, **Enthusiast**** |

| L7 | ****Visual/iconographic**** ? status chips ?/??/?; illustrated capture guide (UI_PROPOSAL A1; VIDEO_RECORDING_GUIDELINES §4; handoff) | ****All**** |

| L8 | ****WhatsApp-friendly share / feedback**** ? easy-to-forward downloaded reel; WhatsApp feedback channel (FLYER; UI_ALPHA_EXECUTION_PLAN; handoff) | ****Enthusiast, Parent**** (share), ****Coach**** (feedback) |

| L9 | ****Background processing, come back later**** ? async jobs; "close the tab" (OVERVIEW; UI_PROPOSAL B1) | ****All**** (slow uplinks, long videos) |

| L10 | ****Effortless correction loop**** ? A4 accept/reject/nudge + ****bulk "all good"**, simple chips** (UI_PROPOSAL A4; OVERVIEW) | ****Coach**** (primary corrector / co-builder) |

| L11 | ****Honest, calm expectation-setting**** ? today-vs-roadmap split; misses = feedback (OVERVIEW) | ****All**** |

| L12 | ****Show-don't-tell**** ? lead with a sample reel of ***their*** court, not accuracy numbers (PLAYBOOK; PMF §4; OVERVIEW) | ****Coach**** (decision-maker), ****Parent**** |

| L13 | ****Plain, reassuring privacy + opt-in + minors-protection**** copy at the right moments (OVERVIEW; PERSONALIZATION_PLAN consent/minors guardrails) | ****Coach**** (records others), ****Parent**** |

| L14 | ****Kannada-first, Hindi-next localization**** with English fallback; ICU plurals; localized errors/report (I18N_PLAN; handoff) | ****Coach, Parent**** (Kannada-comfortable), All |

| L15 | ****Per-player reels**** (coach ? student/parent) as an ****optional overlay****, not baked in (UI_PROPOSAL §2; PERSONALIZATION_PLAN "coach?athlete optional overlay") | ****Coach ? Parent/Enthusiast**** |

| L16 | ****Verdict banner + clear status**** so users know "ready / problem / failed" without reading logs (UI_PROPOSAL A1/A3) | ****All**** |

| L17 | ****Re-process replaces, doesn't duplicate**** ? simple mental model ("re-running replaces its earlier result") (OVERVIEW "How to use it") | ****All**** |

7. Open owner decisions the engineer should know are unresolved

- ****Mobile-perfect layout**** is an explicit alpha ****non-goal**** today (UI_PROPOSAL §3) ? but the non-tech-savvy mandate is mobile-first (handoff). Treat as a known tension to raise, not a settled spec.
- ****Localization owner decisions still open**** (I18N_PLAN §4 / handoff): (1) ****Indic font licensing**** ? recommend OFL-for-fonts exception, self-host Noto Sans Kannada/Devanagari; (2) ****v1 locale set**** = en + kn, hi next, always-fallback en; (3) ****backend-localization depth**** ? customer verdict/report + validation-rejection reasons first, operator text stays English; (4) ****which surface first**** ? marketing `site/` (first-contact friction) vs the `web/` SPA console.
- ****Surface drift to flag:**** I18N_PLAN was written for the ****old static UI**** (`backend/api/static/`); the live product is now the ****`khelsutra` `web/` SPA**** (React/Vite/TS/Tailwind) plus a vanilla marketing ****`site/`****. The framework-agnostic catalog/negotiation design carries over (SPA ? react-i18next; site ? tiny `t()` shim), per the next-session handoff §"THE adaptation."
- ****Implementation is owner-gated.**** CLAUDE.md + UI_PROPOSAL: no platform/UI implementation starts without explicit owner approval; this is a research-then-plan task.

****Relevant file paths (all absolute):****

- `C:\Users\avidu\Projects\badminton-highlight-indexer\docs\UI\_PROPOSAL.md` (personas §2, 6 screens §3)
- `C:\Users\avidu\Projects\badminton-highlight-indexer\docs\KHELSUTRA\_PRODUCT\_OVERVIEW.md` (who it's for, plain-tone "Read this honestly" copy)
- `C:\Users\avidu\Projects\badminton-highlight-indexer\docs\PMF\_FIELD\_SIGNALS.md` and `docs\PMF\_MARKET\_RESEARCH.md` (buyers/users, WhatsApp-native, parents pivot)
- `C:\Users\avidu\Projects\badminton-highlight-indexer\docs\PRODUCT\_FIELD\_ASSOCIATE\_FLYER.md`, `docs\FIELD\_VISIT\_PLAYBOOK.md`, `docs\FIELD\_VISIT\_ANSWER\_SHEET.md` (Kannada/Hindi/English on-

the-ground reality)

- `C:\Users\avidu\Projects\badminton-highlight-indexer\docs\PERSONALIZATION\_PLAN.md`
(coach?athlete optional overlay; no live self-improvement yet)
- `C:\Users\avidu\Projects\badminton-highlight-indexer\docs\UI\_ALPHA\_EXECUTION\_PLAN.md` (UX sequencing + status)
- `C:\Users\avidu\Projects\badminton-highlight-indexer\docs\I18N\_PLAN.md` (Kannada-first localization, owner-gated)
- `C:\Users\avidu\Projects\badminton-highlight-indexer\docs\VIDEO\_RECORDING\_GUIDELINES.md` (the recording-education content to surface)
- `C:\Users\avidu\projects\C--Users-avidu-Projects-badminton-highlight-indexer\memory\next-session-ux-localization.md` (owner directive + the static-UI?SPA/site adaptation)